

Bounded Schrödinger potentials for robust learning in generative models

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September 30, 2026

Abstract

We propose a new class of M-estimators (based on a robust Sinkhorn divergence) for robust learning in generative models and show that their influence function is bounded. This guarantees B-robustness. The standard influence function representation is inapplicable because the estimating equation depends on the entire empirical measure and lacks the usual observation-wise decomposition. We overcome this by computing the Fréchet derivative, in total variation, via a Schrödinger linear-response theorem. Using Hadamard differentiability in the bounded-Lipschitz topology, we establish also asymptotic normality of the estimators, with dimension-free root- n rate. Numerical experiments confirm the estimators' stability under data perturbation.

Bounded influence function, Entropic regularisation, Fréchet derivative, Optimal transport

1 Introduction

Optimal transport has emerged as a powerful tool for data analysis, with the Wasserstein distance $W_p, p \geq 1$ defining the minimum cost of transporting mass between distributions. In machine learning, W_p is fundamental for generative models (Genevay et al., 2018); in statistics, it underpins minimum distance estimation (Bassetti and Regazzini, 2006; Bernton et al., 2019). For high-dimensional data, entropic-regularised optimal transport offers favourable sample complexity and avoids the curse of dimensionality (Genevay et al., 2018; Feydy et al., 2019).

Despite these successes, the optimal transport loss function W_p remains sensitive to (even small) perturbations of the data distribution (specifically to outliers), triggering the research on robust alternatives (Mukherjee et al., 2021; Nietert et al., 2022; Ma et al., 2025). Recently, La Vecchia and Liu (2026) proposed Entropic regularized RoBust Optimal Transport

(E-ROBOT), which combines cost truncation with entropic regularisation to achieve both robustness and dimension-free sample complexity. The resulting divergence $\overline{W}_{\epsilon,\lambda}$ is the focus of this note.

We introduce a novel class of M-estimators obtained by minimising the (loss function) $\overline{W}_{\epsilon,\lambda}$ between the empirical measure $\hat{\mu}_n$ and a parametric generative model μ_θ . To characterise their worst bias over a neighborhood of the model distribution and their resistance to outliers, we derive the influence function (see [Hampel \(1974\)](#), and [Hampel et al. \(1986\)](#), (2.7.7), p. 175.) The first-variation formula and Riesz representation give the integral form of the gradient (Lemma 1); the Banach-space implicit function theorem (for infinite-dimensional objects, the Schrödinger potentials) yields the influence function in terms of a linear operator Γ_{μ_0} (see (4.2)). The main technical challenge is boundedness of Γ_{μ_0} on point masses. To this end, we establish a linear-response theorem (Theorem 3) for the Schrödinger system under a spectral gap condition, which we show is implied by bounded potentials (Doebelin minorisation, as in Lemma 2). This yields B -Robustness (Corollary 4). Moreover, we prove the asymptotic normality of our M-estimators, using the Hadamard derivative and the functional delta method. Our results extend the univariate location W_1 -based analysis of [Bassetti and Regazzini \(2006\)](#) to multivariate E-ROBOT, and, unlike recent Monge-Ampère linearisations ([González-Sanz and Sheng, 2024](#)), our characterisation uses derivatives of Schrödinger potentials. Proofs and pseudo-codes are in the [Supplementary Material](#).

2 Local sensitivity of statistical functionals

To assess robustness of an estimator, we consider its behaviour under small perturbations of the underlying distribution. The classical tool for this is the influence function, which measures the first-order effect of a point-mass contamination. Formally, let $\mathcal{X}$ be a Polish space and $\mathcal{P}(\mathcal{X})$ the space of probability measures and let $\|h\|_{\text{TV}} = \sup_{\|\varphi\|_\infty \leq 1} |\int \varphi dh|$ denote the total variation norm, for the signed measure $h \in \mathcal{M}_0(\mathcal{X})$ (finite total variation, zero total mass). For a statistical functional $T : \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}^k$, the Fréchet derivative at μ in the direction h is $D_\mu T(\mu)$ defined by

$$T(\mu + h) = T(\mu) + D_\mu T(\mu)[h] + o(\|h\|_{\text{TV}}). \quad (2.1)$$

The influence function is obtained by evaluating this derivative at a point-mass contamination: $\text{IF}(z; T, \mu) = D_\mu T(\mu)[\delta_z - \mu]$, where δ_z is the Dirac point mass at z . From (2.1) it follows that $T(\mu + h) - T(\mu)$ can be approximated at the first order by $D_\mu T(\mu)[h]$. Thus, if the influence function is bounded as a function of z , the estimator is B -robust ([Hampel et al., 1986](#)). This defines the notion of stability of statistical functionals over gross-error neighbourhoods containing, for $t \ll 1/2$, $\mu_t = \mu + th$, where $h = \mu - \omega$ and ω being a generic measure. In what follows, we will consider $\omega = \delta_z$, which is the Dirac mass at $z \in \mathcal{X}$ and we call stable any statistical functional T such that small deviations (in total variation norm) from a reference μ

imply bounded changes to the functional. Our goal is to derive the influence function for the M-functional $T(\mu) = \arg \min_{\theta} \overline{W}_{\epsilon, \lambda}(\mu, \mu_{\theta})$. Since T is defined implicitly as the minimiser of a loss, we cannot evaluate its derivative directly. Instead, we apply the implicit function theorem to the estimating equation $\Psi(\mu, \theta) = \nabla_{\theta} \overline{W}_{\epsilon, \lambda}(\mu, \mu_{\theta}) = 0$. This approach, which is in line with [Clarke \(1983\)](#), automatically yields the Fréchet derivative of T and, crucially, the $o(\|h\|_{\text{TV}})$ remainder.

3 The E-ROBOT divergence and its first variation

We recall the basics of E-ROBOT; we refer to [La Vecchia and Liu \(2026\)](#) for more details. Set the cost function $c_{\lambda}(x, y) = \min\{\|x - y\|, 2\lambda\}$ and define $W_{\epsilon, \lambda}(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c_{\lambda} d\pi + \epsilon H(\pi \| \mu \otimes \nu)$, where H is the relative entropy between π and $\mu \otimes \nu$. The debiased divergence is $\overline{W}_{\epsilon, \lambda}(\mu, \nu) = W_{\epsilon, \lambda}(\mu, \nu) - 0.5(W_{\epsilon, \lambda}(\mu, \mu) + W_{\epsilon, \lambda}(\nu, \nu))$. The first variation w.r.t. ν is

$$\frac{\delta}{\delta \nu} \overline{W}_{\epsilon, \lambda}(\mu, \nu)(y) = \psi_{\mu, \nu}(y) - u_{\nu}(y), \quad (3.1)$$

where $\psi_{\mu, \nu}$ is the second Schrödinger potential and u_{ν} the self-coupling potential ([Feydy et al., 2019](#); [La Vecchia and Liu, 2026](#)). From the dual representation of $W_{\epsilon, \lambda}$ it follows that boundedness of c_{λ} implies uniform boundedness of the potentials; see Prop. 3 in [La Vecchia and Liu \(2026\)](#).

4 E-ROBOT M-functional and its influence function

We call generative model a collection of distributions $\{\mu_{\theta} : \theta \in \Theta \subset \mathbb{R}^k\}$, which is a correctly specified parametric family (namely, it contains the true data generating model μ_0) with common dominating measure σ and $f_{\theta} = d\mu_{\theta}/d\sigma$. The score $s_{\theta} = \nabla_{\theta} \log f_{\theta}$ satisfies $\int \|s_{\theta}\| d\mu_{\theta} < \infty$.

For a random sample $(X_1, X_2, \dots, X_n)$, the E-ROBOT estimator is $\hat{\theta}_n = \arg \min_{\theta \in \Theta} \overline{W}_{\epsilon, \lambda}(\hat{\mu}_n, \mu_{\theta})$, with $\hat{\mu}_n = \sum_{i=1}^n \delta_{X_i}/n$. The estimating equation is $\Psi_n(\theta) = \nabla_{\theta} \overline{W}_{\epsilon, \lambda}(\hat{\mu}_n, \mu_{\theta}) = 0$. We notice that Ψ_n depends on the entire empirical measure: this is not of the M-estimator form typically met in robust statistics for multivariate random variables (see e.g. [Hampel et al., 1986](#), Ch. 4.2c), where the estimating function is an average of observation-wise contributions. This prevents the usual influence function representation and requires a different approach. To derive it, let us start from existence, measurability, and consistency of the E-ROBOT M-functional, which all follow from the general minimum-distance theory of [Bernton et al. \(2019\)](#), adapted by replacing their W_p with our $\overline{W}_{\epsilon, \lambda}$. Because $\overline{W}_{\epsilon, \lambda}$ is continuous with respect to weak convergence (thanks to the bounded cost c_{λ}) the arguments of Theorems 2.1 and 2.2 carry over without requiring moment assumptions. In particular, the analogue of Assumption 2.1 of [Bernton et al. \(2019\)](#), namely $\overline{W}_{\epsilon, \lambda}(\hat{\mu}_n, \mu_0) \rightarrow 0$ almost surely, follows from the strong law of large numbers for empirical measures. Continuity of $\theta \mapsto \overline{W}_{\epsilon, \lambda}(\nu, \mu_{\theta})$ follows from the Sinkhorn dual representation and uniform

boundedness of the Schrödinger potentials (La Vecchia and Liu, 2026). Measurability follows from the selection argument in Theorem 2.2 of Bernton et al. (2019), requiring only that $\mathcal{X}$ is Polish, Θ is σ -compact, and the loss is jointly measurable. We refer to Ma et al. (2025) for similar calculation in the ROBOT case. Thus, under standard conditions, the E-ROBOT estimator exists, is measurable, and is consistent for the minimiser of $\theta \mapsto \overline{W}_{\epsilon, \lambda}(\mu_0, \mu_\theta)$, assuming uniqueness. Fisher consistency holds automatically because $\overline{W}_{\epsilon, \lambda}(\mu, \nu) = 0$ if and only if $\mu = \nu$; hence $T(\mu_0) = T(\mu_{\theta_0}) = \theta_0$. The population estimating equation is therefore $\Psi(\mu, \theta) = \nabla_\theta \overline{W}_{\epsilon, \lambda}(\mu, \mu_\theta)$. Let μ_t be the contaminated distribution, and let $\theta_t = T(\mu_t)$. Formally, if the statistical functional is (Fréchet) differentiable and the population Hessian $M(\theta_0) = -\nabla_{\theta\theta}^2 \overline{W}_{\epsilon, \lambda}(\mu_0, \mu_{\theta_0})$ is nonsingular, differentiating the first-order condition $\Psi(\mu_t, \theta_t) = 0$ with respect to t at $t = 0$ yields

$$\text{IF}(z; T, \mu_0) = M(\theta_0)^{-1} \frac{\partial \Psi}{\partial \mu} \Big|_{\mu=\mu_0} (\delta_z - \mu_0), \quad \text{where } \delta_z - \mu_0 \in \mathcal{M}_0(\mathcal{X}). \quad (4.1)$$

We refer to Appendix 1 of the [Supplementary material](#) for this derivation. In Lemma 1 below, we prove that $\Psi(\mu_0, \theta_0)$ admits an integral representation in terms of Schrödinger potential $\psi_{\mu, \mu_{\theta_0}}$

$$\frac{\partial \Psi}{\partial \mu} \Big|_{\mu=\mu_0} (\delta_z - \mu_0) = \Gamma_{\mu_0}(\delta_z - \mu_0) := D_\mu \left[\int \psi_{\mu, \mu_{\theta_0}}(y) s_{\theta_0}(y) \mu_{\theta_0}(dy) \right] \Big|_{\mu=\mu_0} (\delta_z - \mu_0),$$

where the linear operator $\Gamma_{\mu_0} : \mathcal{M}_0(\mathcal{X}) \rightarrow \mathbb{R}^k$. Thus, we have:

$$\text{IF}(z; T, \mu_0) = M(\theta_0)^{-1} \Gamma_{\mu_0}(\delta_z - \mu_0). \quad (4.2)$$

The utility of (4.2) rests on establishing conditions for the Fréchet differentiability and on obtaining an expression for the linear operator Γ_{μ_0} . In the M-estimator setting where the estimator admits (via first-order von Mises expansion, see e.g. La Vecchia et al. (2012)) an observation-wise decomposition, the influence function depends on the estimating function (namely the derivative of the loss function). Similarly, in our E-ROBOT setting, since the loss $\overline{W}_{\epsilon, \lambda}$ depends on the Schrödinger potentials, we anticipate that $\text{IF}(z; T, \mu_0)$ in (4.2) depends on the derivative of the potentials. The linear-response theorem in the next section proves this statement and formalizes the Fréchet derivative calculation (existence and boundedness).

5 Linear-response theorem and B -robustness

First, let us introduce the following

Lemma 1. *For any probability measure μ , $\Psi(\mu, \theta_0) = \int (\psi_{\mu, \mu_{\theta_0}}(y) - u_{\mu_{\theta_0}}(y)) s_{\theta_0}(y) \mu_{\theta_0}(dy)$, and the function $\psi_{\mu, \mu_{\theta_0}} - u_{\mu_{\theta_0}}$ is uniformly bounded, with constants depending only on λ and ϵ .*

The result follows from the Riesz representation theorem for Ψ , while the bounded density allows integration with respect to the score. The influence function requires the derivative of Ψ with respect to μ_0 . When differentiating Ψ with respect to the first marginal,

the term $u_{\mu_{\theta_0}}$ drops out because it depends only on the second marginal μ_{θ_0} , which is held fixed. Then, define the (Schrödinger) map $\mathcal{S} : (\mu, \nu) \mapsto (\phi_{\mu, \nu}, \psi_{\mu, \nu})$, with potentials solving the dual problem with cost c_λ . Let $K(x, y) = e^{-c_\lambda(x, y)/\epsilon}$, bounded away from zero and infinity. For a base pair (μ_0, ν_0) , let $(\phi_0, \psi_0) = \mathcal{S}(\mu_0, \nu_0)$ and denote the optimal coupling $\pi_0(dx, dy) = e^{\phi_0(x) + \psi_0(y)} K(x, y) \mu_0(dx) \nu_0(dy)$. The conditional kernels are $P_x(dy) = \pi_0(dy|x)$, and $Q_y(dx) = \pi_0(dx|y)$. Define the operator E by $E\varphi(x) = \int \varphi(y) P_x(dy)$, and its adjoint E^* by $E^*\varphi(y) = \int \varphi(x) Q_y(dx)$. The key technical condition needed is the invertibility of $I - E^*E$ on the space of mean-zero bounded functions $L_0^\infty(\mu_0) = \{f \in L^\infty(\mu_0) : \int f d\mu_0 = 0\}$. The invertibility follows from the spectral gap result, proved in:

Lemma 2. *Let μ_0 be any probability measure with strictly positive density. Consider the self-coupling optimal Schrödinger plan $\pi_0(dx, dy)$ between μ_0 and itself, and let $P_x(dy) = \pi_0(dy|x)$. Under the bounded-potential property, the operator E satisfies the L^∞ contraction $\|Ef\|_{L^\infty(\mu_0)} \leq (1 - m)\|f\|_{L^\infty(\mu_0)}$, $f \in L^\infty(\mu_0)$, with $0 < m < 1$. The same holds for E^* , hence*

$$\|E^*E\|_{L_0^\infty(\mu_0) \rightarrow L_0^\infty(\mu_0)} < 1.$$

Thanks to Lemma 2, we can prove the following

Theorem 3. *Let μ_0 be a probability measure with strictly positive density, and consider the self-coupling base pair (μ_0, μ_0) , with optimal Schrödinger plan $\pi_0(dx, dy)$ between μ_0 and itself. The Schrödinger map $\mathcal{S}$ is Fréchet differentiable at (μ_0, μ_0) as a map from $\mathcal{M}_0(\mathcal{X})^2$ to $L^\infty(\mathcal{X})^2$. For given signed measures $h_\mu, h_\nu \in \mathcal{M}_0(\mathcal{X})$, let $(\dot{\phi}, \dot{\psi})$ denote the derivative of $\mathcal{S}$ at (μ_0, μ_0) in the direction (h_μ, h_ν) . It satisfies the linearised system*

$$\begin{cases} \dot{\phi}(x) + \int P_x(dy) \dot{\psi}(y) = - \int p_x(y) h_\nu(dy), \\ \dot{\psi}(y) + \int Q_y(dx) \dot{\phi}(x) = - \int q_y(x) h_\mu(dx), \end{cases} \quad (5.1)$$

$$\quad (5.2)$$

where the Schrödinger potentials are normalized so that $\int \dot{\phi} d\mu_0 = 0$, where $P_x(dy) = \pi_0(dy | x)$, $Q_y(dx) = \pi_0(dx | y)$, and $p_x(y) = dP_x/d\mu_0(y)$, $q_y(x) = dQ_y/d\mu_0(x)$ are bounded conditional densities with respect to μ_0 . Moreover, there exists a constant $C < \infty$, depending on ϵ, λ , such that for all h_μ, h_ν with $\|h_\mu\|_{TV}, \|h_\nu\|_{TV} \leq 2$, $\|\dot{\phi}\|_{L^\infty(\mathcal{X})} + \|\dot{\psi}\|_{L^\infty(\mathcal{X})} \leq C$.

To compute the influence function, we need the derivative of the map $\mu \mapsto \psi_{\mu, \mu_{\theta_0}}$. Crucially, the second marginal μ_{θ_0} is held fixed when differentiating with respect to μ , so we set $h_\nu = 0$ in the linearised system (A.4)–(A.5) of Theorem 3. For point-mass perturbations such as $h_\mu = \delta_z - \mu_0$, the system has solution $(\dot{\phi}_z, \dot{\psi}_z)$. The second component gives the derivative of the second potential with respect to the first marginal: $\dot{\psi}_z(y) := D_\mu \psi_{\mu, \mu_{\theta_0}}[\delta_z - \mu_0](y)|_{\mu=\mu_0}$, which is the quantity required in the definition of Γ_{μ_0} . To make the connection explicit, note that Γ_{μ_0} is defined as a Fréchet derivative of an integral with respect to the fixed measure μ_{θ_0} . To bring the

derivative inside the integral, we apply the dominated convergence theorem to the derivative of the integrand $y \mapsto \psi_{\mu, \mu_{\theta_0}}(y) s_{\theta_0}(y)$. Theorem 3 provides the pointwise derivative $\dot{\psi}_z(y)$, and the uniform bound $\|\dot{\psi}_z\|_{L^\infty} < \infty$ combined with the score integrability $\int \|s_{\theta_0}\| d\mu_{\theta_0} < \infty$ yields an integrable dominating function. Hence,

$$\Gamma_{\mu_0}(\delta_z - \mu_0) = \int D_\mu \psi_{\mu, \mu_{\theta_0}}[\delta_z - \mu_0](y) s_{\theta_0}(y) \mu_{\theta_0}(dy) = \int \dot{\psi}_z(y) s_{\theta_0}(y) \mu_{\theta_0}(dy). \quad (5.3)$$

This is in line with the usual notion of influence function: $\dot{\psi}_z$ is the infinitesimal change in the cross-coupling potential under a point-mass contamination of the first marginal at z . Using (4.2) and (A.3), one can prove $\mathbb{E}_{\mu_0}[\text{IF}(Z; T, \mu_0)] = 0$; see Remark 1 in the [Supplementary Material](#). Moreover, from Lemma 2, for $h_\mu = \delta_z - \mu_0$ with $h_\nu = 0$, we have that $B_z(y) = -\int q_y(x)(\delta_z - \mu_0)(dx)$ is uniformly bounded in L^∞ over all z , because $0 < m \leq q_y(x) \leq \bar{M} < \infty$. Hence $\dot{\psi}_z = (I - E^*E)^{-1}B_z$ satisfies $\sup_{z \in \mathcal{X}} \|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} < \infty$. So, since $\|\delta_z - \mu_0\|_{\text{TV}} \leq 2$ uniformly in z , the uniform bound on $\dot{\psi}_z$ (Theorem 3) gives $\sup_z \|\Gamma_{\mu_0}(\delta_z - \mu_0)\| \leq \sup_z \|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} \int \|s_{\theta_0}(y)\| \mu_{\theta_0}(dy) \leq \sup_z \|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} \|\nabla_{\theta_0} \mu_{\theta_0}\|_{\text{TV}} < \infty$. Thus, we state:

Corollary 4. *The E-ROBOT M-estimator is B-robust (bounded influence function).*

6 Numerical examples

The pseudocode to implement our method is available in the [Supplementary Material](#). The routines to reproduce our experiments are available at https://github.com/dvdlvc/E-ROBOT/tree/main/Git_Code_M_EROBOT.

6.1 Location model

Gaussian case. We consider the multivariate Gaussian location model $\mu_\theta = \mathcal{N}(\theta, I_d)$, $\theta \in \mathbb{R}^d$, with unbounded (component-wise) score $s_\theta(x) = x - \theta$; the MLE is not B -robust. The boundedness of the conditional density (Lemma 2) gives the spectral gap even when $\mathcal{X} = \mathbb{R}^2$, so our theory implies that the E-ROBOT estimator is B -robust. To illustrate numerically the implied stability of estimates, we set $d = 2$, $n = 200$, $\theta_0 = (0, 0)$, $\epsilon = 1$, $\lambda = 2.5$ (which controls the bound on the potentials, see [Ma et al. \(2025\)](#)), and outliers at 4 with contamination $t = 0, 0.05, 0.15$. Figure 1 displays violin plots (200 Monte Carlo runs) and sensitivity curves. The E-ROBOT estimator remains stable under contamination (smaller bias than the MLE for $t > 0$) while preserving a distribution similar to the MLE at $t = 0$. The influence function of the MLE is unbounded, whereas the sensitivity curves for θ_1 are not only bounded but also smoothly (thanks to the entropic regularization term) redescending (thanks to the of truncation of the cost function), decaying to zero for $|z| > 5$. This assigns zero weight to observations which lie far out.

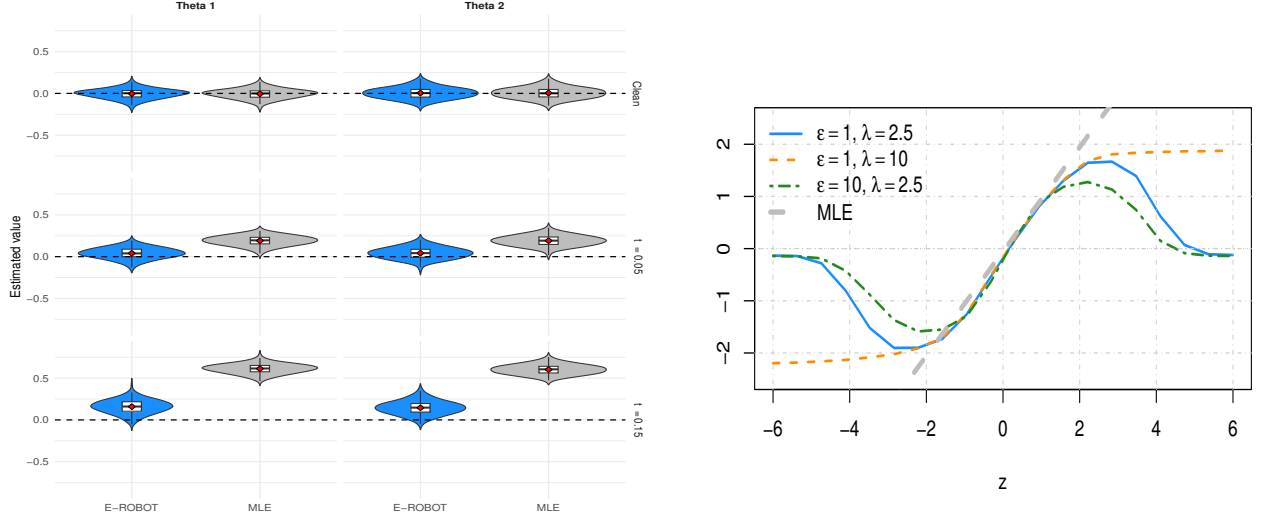


Figure 1: E-ROBOT M-estimator vs MLE for bivariate Gaussian model. Left: violin plots; right: sensitivity curves, for different (λ, ϵ) and contamination value Z .

t_2 -distribution case. We now assess the E-ROBOT M-estimator under heavy-tailed data, using a t_2 distribution. Thanks to its connection to ROBOT via the bounded c_λ , the E-ROBOT estimator does not require finite moments and can be computed also when moments do not exist; see (Ma et al., 2025). For the t_2 location model, the MLE is known to be redescending and hence B -Robust (Maronna et al., 2006, Ch. 2). This comparison is informative about the trade-off between robustness and efficiency, when the sample size is small. We consider dimensions $d = 2$ and $d = 4$, contamination levels $\eta = 0, 0.05, 0.10$, and outliers at -20 in each dimension. Table 1 reports the MSE and absolute bias for both estimators. The E-ROBOT estimator exhibits similar bias and MSE to the MLE across all contamination levels, confirming its B -robustness with no large efficiency loss at $t = 0$.

Estimator	t	$d = 2$		$d = 4$	
		MSE	Abs Bias	MSE	Abs Bias
E-ROBOT	0	0.066	0.147	0.107	0.129
E-ROBOT	5%	0.070	0.149	0.120	0.139
E-ROBOT	10%	0.062	0.141	0.119	0.138
MLE	0	0.050	0.126	0.094	0.120
MLE	5%	0.050	0.125	0.097	0.123
MLE	10%	0.049	0.128	0.098	0.125

Table 1: Comparison of MLE and E-ROBOT for d -ivariate t_2 location ($n = 60$). MSE is the mean squared Error, while Abs Bias is the mean absolute error across components.

Figure 2 shows the violin plots for the first and second components of θ in dimension $d = 4$ at $t = 0\%$ and $t = 10\%$ (the plots for the other components are similar). The two estimators have very comparable distributions at $t = 0\%$ and both remain stable under contamination.

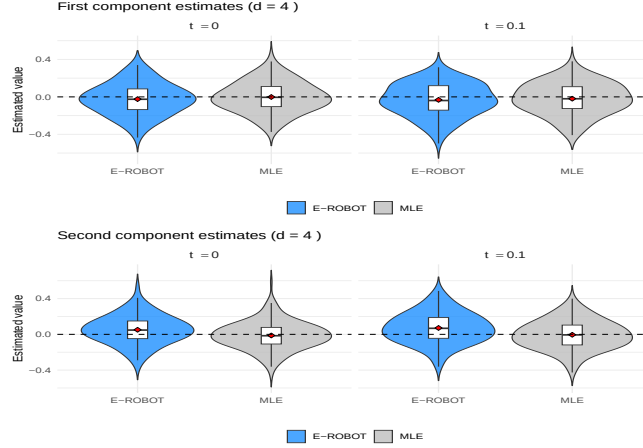


Figure 2: E-ROBOT vs MLE for t_2 , $d = 4$: violin plots for the first (top) and second (bottom) components of θ .

6.2 Generative adversarial network

We now consider a popular machine learning problem: a generative model where θ represents the parameter of a deep neural network. We adopt the two-dimensional ring distribution, a standard benchmark in the literature—note, however, that $\overline{W}_{\epsilon, \lambda}$ can be computed for larger d , as illustrated numerically in [La Vecchia and Liu \(2026\)](#). The generator $G_\theta : \zeta \rightarrow \mathcal{X}$ maps latent (Gaussian) noise $\zeta \sim P_\zeta$ to data $X = G_\theta(\zeta)$. The objective is to minimise $\overline{W}_{\epsilon, \lambda}$ between the pushforward $G_{\theta\#}P_\zeta$ and the data distribution. To train the generator, we proceed as in https://pythonot.github.io/auto_examples/backends/plot_wass2_gan_torch.html: we use batches of size $n = 400$ and 200 iterations—at each iteration, we sample n fresh data from the ring distribution and fresh latent variables from P_ζ to generate synthetic samples of size n . The true data are generated as $X = (\cos U, \sin U) + \sigma w$, with $U \sim \text{Unif}[0, 2\pi]$, $w \sim \mathcal{N}(0, I_2)$, and $\sigma = 0.1$. The generator is a multilayer perceptron with hidden layers of sizes 200 and 500 and a total of more than 100000 parameters. We compare: (1) the exact W_2 distance (computed via the `Python` routine `ot.emd2`), which is based on an unbounded cost; (2) the E-ROBOT loss $\overline{W}_{\epsilon, \lambda}$, which adds cost truncation and debiasing. For E-ROBOT, we set $\epsilon = 1$ and $\lambda = 0.3$. In the contaminated scenario, $t = 5\%$ of the data are replaced by outliers from $\mathcal{N}(3, 0.1^2 I_2)$. Figure 3 displays the loss curves: the key distinction is that the W_2 loss is affected by contamination (the curves with and without contamination are shifted), whereas the E-ROBOT curve is stable (almost no shift due to contamination), demonstrating its robustness. The final generated samples (Figure 3) show that under clean data both methods recover the ring. In contrast, under contamination, the W_2 generator is pulled towards the outliers, while E-ROBOT correctly ignores them and recovers the true ring distribution.

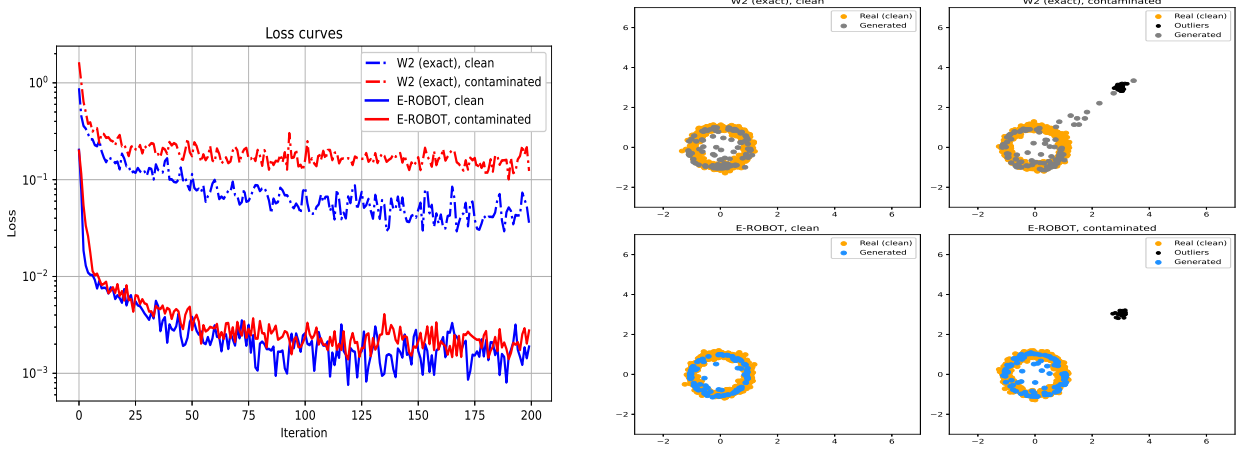


Figure 3: E-ROBOT M-estimator vs W_2 for GAN: ring distribution. Left panel is for loss curves. Right panel: actual and generated data.

7 Further asymptotic developments

The derived influence function plays a role in further asymptotic developments. To illustrate one of them, we consider the asymptotic distribution of our estimators. With this regard we mention that, for smooth cost functions, Goldfeld et al. (2024) establish (in the bounded-Lipschitz topology) first-order Hadamard differentiability of (a map depending on) the entropic optimal transport potentials. In Proposition 1 of the Supplementary Material, we explain how to adapt the result to the map $\Psi(\mu_0, \theta_0)$, defined in the E-ROBOT setting where c_λ is not smooth. Although Hadamard differentiability is weaker than the Fréchet differentiability established in Theorem 3, it has a key role in the functional delta method (van der Vaart and Wellner, 1996), as it is compatible with the weak convergence of $\sqrt{n}(\mu_n - \mu_0)$. This allows us to prove the asymptotic normality of our M-estimators, assuming that: either (a) $\mathcal{X}$ is compact; or (b) $\mathcal{X} = \mathbb{R}^d$ and μ_0 satisfies a condition sufficient for the unit ball of bounded 1-Lipschitz functions on $\mathcal{X}$ to be μ_0 -Donsker. See the Supplementary Material for a discussion on this Donsker property. Thus, we state

Theorem 5. *Under the Hadamard differentiability of $\Psi(\mu, \theta_0)$ at μ_0 in the bounded-Lipschitz topology, the Donsker property, and the standard M-estimation regularity conditions on $\theta \mapsto \Psi(\mu_0, \theta)$, $\hat{\theta}_n$ satisfies $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{L}} \mathcal{N}(0, V(\theta_0))$, with $V(\theta_0) = \text{Var}(\text{IF}(Z; T, \mu_0))$, $Z \sim \mu_0$.*

The rate does not depend on the dimension d . This opens the door to the investigation of dimension-free higher-order robustness, which can define robust tests and saddlepoint techniques in a neighbourhood of μ_0 (Heritier and Ronchetti, 1994; La Vecchia et al., 2012). It also paves the way for differentially private generative models, where the bounded influence function (Avella-Medina (2021)) of E-ROBOT M-estimators may play a key role.

A Supplementary Material

In this supplementary material, we provide the proofs of Lemmas and Theorems stated in the main paper. Moreover, we give the pseudo code needed to implement our method. In Appendix A and Appendix B we provide some additional detailed calculations.

A.1 Preliminary results

In what follows we will need a result on boundedness of several quantities. In this section, we prove these preliminary results (uniform bound and Lipschitz continuity of the potentials) that we will use later on in the proofs of Lemma 1, Lemma 2 and Lemma 3.

Let μ, ν be probability measures on a Polish space $\mathcal{X}$, and let (ϕ, ψ) be the optimal Schrödinger potentials for the E-ROBOT cost c_λ with regularization $\epsilon > 0$, normalized by $\int \phi d\mu = 0$. Then there exist constants $C_0 = C_0(\lambda)$ and $\kappa_0 = \kappa_0(\epsilon, \lambda)$, not depending on μ, ν or the dimension of $\mathcal{X}$, such that

$$\|\phi\|_\infty \leq C_0, \quad \|\psi\|_\infty \leq C_0, \quad |\phi(x) - \phi(x')| \leq \kappa_0 |x - x'|, \quad |\psi(y) - \psi(y')| \leq \kappa_0 |y - y'|, \quad (\text{A.1})$$

for all $x, x', y, y' \in \mathcal{X}$. This completes the results in Proposition 3 of [La Vecchia and Liu \(2026\)](#), providing explicit expressions for the bounds and the constants C_0 and κ_0 .

To prove (A.1), let $K(x, y) = e^{-c_\lambda(x, y)/\epsilon}$. Since c_λ is bounded by 2λ , we have

$$m_K := e^{-2\lambda/\epsilon} \leq K(x, y) \leq 1,$$

and because c_λ is 1-Lipschitz,

$$|K(x, y) - K(x', y)| \leq \frac{1}{\epsilon} |x - x'|, \quad |K(x, y) - K(x, y')| \leq \frac{1}{\epsilon} |y - y'|.$$

The optimal potentials satisfy the Schrödinger fixed-point equations

$$\phi(x) = -\epsilon \log \int e^{\psi(y)/\epsilon} K(x, y) d\nu(y), \quad \psi(y) = -\epsilon \log \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x). \quad (\text{A.2})$$

After the normalization $\int \phi d\mu = 0$, there exists $C_0 = C_0(\lambda)$ such that $\|\phi\|_\infty \leq C_0$ and $\|\psi\|_\infty \leq C_0$.

To see this, recall that the entropic optimal transport cost satisfies

$$W_{\epsilon, \lambda}(\mu, \nu) \leq 2\lambda,$$

because the independent coupling $\mu \otimes \nu$ yields cost $\int c_\lambda d(\mu \otimes \nu) \leq 2\lambda$ and relative entropy zero.

For optimal potentials, the dual representation gives

$$W_{\epsilon,\lambda}(\mu, \nu) = \int \phi d\mu + \int \psi d\nu.$$

Since $\int \phi d\mu = 0$, we get

$$\int \psi d\nu = W_{\epsilon,\lambda}(\mu, \nu) \leq 2\lambda.$$

Now, from the first equation in (A.2),

$$e^{-\phi(x)/\epsilon} = \int e^{\psi(y)/\epsilon} K(x, y) d\nu(y) \geq m_K \int e^{\psi(y)/\epsilon} d\nu(y).$$

By Jensen's inequality,

$$\int e^{\psi/\epsilon} d\nu \geq e^{\int \psi d\nu/\epsilon} \geq e^{-2\lambda/\epsilon},$$

so

$$e^{-\phi(x)/\epsilon} \geq m_K e^{-2\lambda/\epsilon} = e^{-4\lambda/\epsilon},$$

which gives $\phi(x) \leq 4\lambda$. From the first equation again,

$$e^{-\phi(x)/\epsilon} = \int e^{\psi/\epsilon} K(x, y) d\nu(y) \leq \int e^{\psi/\epsilon} d\nu(y).$$

We need an upper bound for $\int e^{\psi/\epsilon} d\nu$. From the second equation,

$$e^{-\psi(y)/\epsilon} = \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x) \geq m_K \int e^{\phi/\epsilon} d\mu.$$

Thus

$$e^{\psi(y)/\epsilon} \leq \frac{1}{m_K \int e^{\phi/\epsilon} d\mu}.$$

Because $\int \phi d\mu = 0$, Jensen gives $\int e^{\phi/\epsilon} d\mu \geq 1$. Hence

$$e^{\psi(y)/\epsilon} \leq \frac{1}{m_K} = e^{2\lambda/\epsilon},$$

so

$$\psi(y) \leq 2\lambda.$$

We now derive a lower bound for ψ . Since we have already shown $\phi(x) \leq 4\lambda$, we get

$$\int e^{\phi/\epsilon} d\mu \leq e^{4\lambda/\epsilon}.$$

Therefore, from the second equation in (A.2),

$$e^{-\psi(y)/\epsilon} = \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x) \leq e^{4\lambda/\epsilon},$$

which gives

$$\psi(y) \geq -4\lambda.$$

Thus we have shown

$$-2\lambda \leq \phi(x) \leq 4\lambda, \quad -4\lambda \leq \psi(y) \leq 2\lambda,$$

so taking $C_0 = 4\lambda$ bounds both potentials uniformly.

Next, we prove Lipschitz continuity. Let $C_0 = 4\lambda$ be the sup-norm. For any $x, x' \in \mathcal{X}$, from (A.2),

$$|\phi(x) - \phi(x')| \leq \epsilon \left| \log \frac{\int e^{\psi/\epsilon} K(x, y) d\nu(y)}{\int e^{\psi/\epsilon} K(x', y) d\nu(y)} \right|.$$

Because the integrals are bounded below by $m_K e^{-C_0/\epsilon}$ and above by $e^{C_0/\epsilon}$, the mean value theorem applied to the logarithm on this interval gives the general bound

$$|\log a - \log b| \leq \frac{|a - b|}{\min(a, b)} \leq \frac{e^{C_0/\epsilon}}{m_K} |a - b| \quad \text{for all } a, b \in [m_K e^{-C_0/\epsilon}, e^{C_0/\epsilon}].$$

Specialising this to

$$a = \int e^{\psi/\epsilon} K(x, y) d\nu(y), \quad b = \int e^{\psi/\epsilon} K(x', y) d\nu(y)$$

yields

$$\left| \log \frac{\int e^{\psi/\epsilon} K(x, y) d\nu(y)}{\int e^{\psi/\epsilon} K(x', y) d\nu(y)} \right| \leq \frac{e^{C_0/\epsilon}}{m_K} \left| \int e^{\psi/\epsilon} K(x, y) d\nu(y) - \int e^{\psi/\epsilon} K(x', y) d\nu(y) \right|.$$

Moreover,

$$\left| \int e^{\psi/\epsilon} K(x, y) d\nu(y) - \int e^{\psi/\epsilon} K(x', y) d\nu(y) \right| \leq e^{C_0/\epsilon} \frac{1}{\epsilon} |x - x'|.$$

Combining the two bounds yields

$$|\phi(x) - \phi(x')| \leq \epsilon \frac{e^{C_0/\epsilon}}{m_K} e^{C_0/\epsilon} \frac{1}{\epsilon} |x - x'| = \frac{e^{2C_0/\epsilon}}{m_K} |x - x'|.$$

Therefore we may take

$$\kappa_0 := \frac{e^{2C_0/\epsilon}}{m_K} = e^{(2C_0+2\lambda)/\epsilon} = e^{10\lambda/\epsilon},$$

using $C_0 = 4\lambda$ and $m_K = e^{-2\lambda/\epsilon}$. Thus, there exists κ_0 depending only on C_0, ϵ, m_K such that

$$|\phi(x) - \phi(x')| \leq \kappa_0 |x - x'|.$$

A completely symmetric argument gives the same bound for ψ .

A.2 Proof of Lemma 1

By the chain rule, the derivative of the loss functional with respect to the parameter is

$$\Psi(\mu, \theta_0) = D_{\mu_{\theta_0}} \bar{W}_{\epsilon, \lambda}(\mu, \mu_{\theta_0})[\nabla_{\theta_0} \mu_{\theta_0}].$$

From Eq. (2) in the main text, the first variation admits the density $y \mapsto \psi_{\mu, \mu_{\theta_0}}(y) - u_{\mu_{\theta_0}}(y)$. By Lemma 7 (Step 1), the Schrödinger potentials for the pair (μ, μ_{θ_0}) are uniformly bounded in $L^\infty(\mathcal{X})$, with constants depending only on λ and ϵ . Hence the difference $\psi_{\mu, \mu_{\theta_0}} - u_{\mu_{\theta_0}}$ is uniformly bounded. Therefore, by the Riesz representation theorem for the dual of $\mathcal{M}(\mathcal{X})$ equipped with the total variation norm, the action of the derivative on a signed measure h is represented by integration:

$$D_{\mu_{\theta_0}} \bar{W}_{\epsilon, \lambda}(\mu, \mu_{\theta_0})[h] = \int (\psi_{\mu, \mu_{\theta_0}}(y) - u_{\mu_{\theta_0}}(y)) h(dy).$$

Applying this componentwise with $h = \nabla_{\theta_0} \mu_{\theta_0}$ yields

$$\Psi(\mu, \theta_0) = \int (\psi_{\mu, \mu_{\theta_0}}(y) - u_{\mu_{\theta_0}}(y)) \nabla_{\theta_0} \mu_{\theta_0}(dy).$$

Substituting $\nabla_{\theta_0} \mu_{\theta_0} = s_{\theta_0} \mu_{\theta_0}$ gives the result.

A.3 Proof of Lemma 2, Theorem 1, and Corollary 1.

Proof of Lemma 2. Since the E-ROBOT cost is bounded, we have $0 < e^{-2\lambda/\epsilon} \leq K(x, y) \leq 1$. By Proposition 3 of [La Vecchia and Liu \(2026\)](#) and (A.1), there exists a constant $C > 0$ such that $|\phi_0(x)| \leq C$ and $|\psi_0(y)| \leq C$ for all x, y . Therefore,

$$e^{-2C} e^{-2\lambda/\epsilon} \leq e^{\phi_0(x) + \psi_0(y)} K(x, y) \leq e^{2C}.$$

Set

$$D(x) = \int e^{\phi_0(x) + \psi_0(y')} K(x, y') \mu_0(dy'),$$

the same bounds give

$$e^{-2C} e^{-2\lambda/\epsilon} \leq D(x) \leq e^{2C}.$$

Hence the conditional density

$$p_x(y) = \frac{dP_x}{d\mu_0}(y) = \frac{e^{\phi_0(x) + \psi_0(y)} K(x, y)}{D(x)}$$

satisfies

$$0 < m \leq p_x(y) \leq \overline{M} < \infty,$$

with

$$0 < m = e^{-4C} e^{-2\lambda/\epsilon} \quad \text{and} \quad \overline{M} = e^{4C} e^{2\lambda/\epsilon}. \quad (\text{A.3})$$

Thus, for every measurable set A ,

$$P_x(A) = \int_A p_x(y) \mu_0(dy) \geq m \mu_0(A).$$

This is Doeblin's minorisation condition: there exists a Markov kernel R_x such that

$$P_x(\cdot) = m \mu_0(\cdot) + (1 - m) R_x(\cdot).$$

For $f \in L_0^\infty(\mu_0)$, we have $\int f d\mu_0 = 0$, so

$$Ef(x) = \int f(y) P_x(dy) = (1 - m) \int f(y) R_x(dy).$$

Since R_x is a Markov kernel, $\|R_x f\|_{L^\infty} \leq \|f\|_{L^\infty}$, and hence $\|Ef\|_{L^\infty(\mu_0)} \leq (1 - m) \|f\|_{L^\infty(\mu_0)}$. The same argument applies to the reverse kernel Q_y , since $q_y(x)$ satisfies the same uniform bounds, namely $m \leq q_y(x) \leq \overline{M}$. Thus E^* satisfies the same L^∞ contraction with the same factor. Moreover, if $f \in L_0^\infty(\mu_0)$, then

$$\int Ef d\mu_0 = \iint f(y) P_x(dy) \mu_0(dx) = \int f(y) \mu_0(dy) = 0,$$

because μ_0 is the marginal of the self-coupling π_0 . Similarly, $\int E^* f d\mu_0 = 0$, hence E and E^* leave $L_0^\infty(\mu_0)$ invariant. In what follows, for the ease-of-notation we write $\|E\|_{L_0^\infty(\mu_0)}$ in lieu of $\|E\|_{L_0^\infty(\mu_0) \rightarrow L_0^\infty(\mu_0)}$, similarly for E^* and for the combined operator E^*E . Therefore,

$$\|E^*E\|_{L_0^\infty(\mu_0)} \leq \|E\|_{L_0^\infty(\mu_0)} \|E^*\|_{L_0^\infty(\mu_0)} \leq (1 - m)^2 < 1,$$

which implies that the Neumann series $(I - E^*E)^{-1} = \sum_{k=0}^\infty (E^*E)^k$ converges in operator norm on $L_0^\infty(\mu_0)$.

Equipped with Lemma 2, we can prove the linear-response result in Theorem 1. Specifically, we prove that the Schrödinger map $\mathcal{S} : (\mu, \nu) \mapsto (\phi_{\mu, \nu}, \psi_{\mu, \nu})$ is Fréchet differentiable as a map from $\mathcal{M}_0(\mathcal{X})^2$ to $L^\infty(\mathcal{X})^2$. For given signed measures $h_\mu, h_\nu \in \mathcal{M}_0(\mathcal{X})$, let $(\dot{\phi}, \dot{\psi})$ denote the

derivative. It satisfies the linearised system

$$\begin{cases} \dot{\phi}(x) + \int P_x(dy) \dot{\psi}(y) = - \int p_x(y) h_\nu(dy), \end{cases} \quad (\text{A.4})$$

$$\begin{cases} \dot{\psi}(y) + \int Q_y(dx) \dot{\phi}(x) = - \int q_y(x) h_\mu(dx), \end{cases} \quad (\text{A.5})$$

with $\int \dot{\phi} d\mu_0 = 0$, where $p_x(y) = dP_x/d\mu_0(y)$ and $q_y(x) = dQ_y/d\mu_0(x)$ are bounded conditional densities. Moreover, there exists a constant $C < \infty$, depending on ϵ, λ , such that for all h_μ, h_ν with $\|h_\mu\|_{\text{TV}}, \|h_\nu\|_{\text{TV}} \leq 2$,

$$\|\dot{\phi}\|_{L^\infty(\mathcal{X})} + \|\dot{\psi}\|_{L^\infty(\mathcal{X})} \leq C.$$

The self-coupling case is the only one needed for the influence function computation of Section 4. Indeed, by Fisher consistency (La Vecchia and Liu, 2026, Proposition 7), the base pair at which the derivative of $\mathcal{S}$ is evaluated is always $(\mu_0, \mu_{\theta_0}) = (\mu_0, \mu_0)$. The following proof formalizes these statements.

Proof of Theorem 1. The proof consists of two logical parts.

In Step 1, we prove that the operator $\mathcal{F}$ is continuously Fréchet differentiable with respect to the potentials (ϕ, ψ) and that its derivative satisfies the invertibility condition required by the Banach-space implicit function theorem. The theorem then yields the existence of the Fréchet derivative of the Schrödinger map $\mathcal{S}$ with respect to the marginal measures (μ, ν) . This gives an abstract expression, that does not directly provide the linearised equations or the uniform bounds needed for B -robustness; these are derived in the next step.

In Step 2, we perform an explicit differentiation of the Schrödinger system along measure perturbations to obtain the linearised equations that characterise this measure-space derivative. Solving these equations with the resolvent $(I - E^*E)^{-1}$ yields the uniform bound in the total variation norm.

After the proof of the Theorem 1, we specialise this derivative to point-mass perturbations $h_\mu = \delta_z - \mu_0$ to derive the form of the influence function as it is expressed in Eq. (7) in Section 5 of the main paper and prove its boundedness—namely, its B -robustness.

Step 1. We work on the closed subspace $\mathcal{B}_0 = \{(\phi, \psi) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{X}) : \int \phi d\mu_0 = 0\}$. On this subspace the constant-shift degeneracy is removed, and the Schrödinger system has a unique solution—we recall that Schrödinger potentials are defined up to an additive constant; e.g. see La Vecchia and Liu (2026), the discussion immediately after their equation (2). The system can be written as a fixed-point equation

$$(\phi, \psi) = \mathcal{F}(\phi, \psi; \mu, \nu)$$

on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{X})$, where

$$\mathcal{F}(\phi, \psi; \mu, \nu) = \begin{pmatrix} -\epsilon \log \int e^{\psi(y) - c_\lambda(x, y)/\epsilon} \nu(dy) \\ -\epsilon \log \int e^{\phi(x) - c_\lambda(x, y)/\epsilon} \mu(dx) \end{pmatrix}. \quad (\text{A.6})$$

In Appendix B, we show that the Fréchet derivative w.r.t. the potentials of $\mathcal{F}$ is continuous in the operator norm. By Lemma 2, the Fréchet derivative of the map

$$(\phi, \psi) \mapsto (\phi, \psi) - \mathcal{F}(\phi, \psi; \mu, \nu)$$

with respect to (ϕ, ψ) is invertible at the self-coupling point $(\phi_0, \psi_0) = (\phi_{\mu_0, \mu_0}, \psi_{\mu_0, \mu_0})$. Thus, the hypotheses of the Banach-space implicit function theorem (see e.g. [1], Ch. X, p.270) are verified: this yields Fréchet differentiability of the Schrödinger map $\mathcal{S}$ with respect to the marginal measures (μ, ν) . This is an infinitesimal approach, as in Hampel et al. (1986): in our derivation the implicit function theorem yields Fréchet differentiability in a neighbourhood of (μ_0, μ_0) with respect to the total variation norm.

Step 2. We derive the explicit equations satisfied by the Fréchet derivative of the Schrödinger map and obtain a uniform bound for the corresponding solutions, proving that $\|\dot{\phi}\|_{L^\infty(\mathcal{X})} + \|\dot{\psi}\|_{L^\infty(\mathcal{X})} \leq C$. To this end, consider perturbations of the marginals

$$\mu_t = \mu_0 + t h_\mu, \quad \nu_t = \nu_0 + t h_\nu,$$

with $h_\mu, h_\nu \in \mathcal{M}_0(\mathcal{X})$, and write the corresponding potentials as

$$\phi_t = \phi_0 + t \dot{\phi} + o(t), \quad \psi_t = \psi_0 + t \dot{\psi} + o(t).$$

Differentiating (in t) the first component of the fixed-point equation,

$$\phi_t(x) = -\epsilon \log \int e^{\psi_t(y) - c_\lambda(x, y)/\epsilon} \nu_t(dy),$$

and evaluating the derivative at $t = 0$ gives

$$\dot{\phi}(x) = -\frac{\int e^{\psi_0(y) - c_\lambda(x, y)/\epsilon} \dot{\psi}(y) \nu_0(dy)}{\int e^{\psi_0(y) - c_\lambda(x, y)/\epsilon} \nu_0(dy)} - \frac{\int e^{\psi_0(y) - c_\lambda(x, y)/\epsilon} h_\nu(dy)}{\int e^{\psi_0(y) - c_\lambda(x, y)/\epsilon} \nu_0(dy)}. \quad (\text{A.7})$$

To see this, let

$$G(t) := \int e^{\frac{\psi_t(y) - c_\lambda(x, y)}{\epsilon}} \nu_t(dy)$$

so $\phi_t(x) = -\epsilon \log G(t)$ and $\dot{\phi}(x) = -\epsilon G'(0)/G(0)$. Using the linear expansion $\nu_t = \nu_0 + t h_\nu + o(t)$,

the dominated convergence theorem allows us to compute

$$G'(0) = \frac{1}{\epsilon} \int e^{\frac{\psi_0(y) - c_\lambda(x,y)}{\epsilon}} \dot{\psi}(y) \nu_0(dy) + \int e^{\frac{\psi_0(y) - c_\lambda(x,y)}{\epsilon}} h_\nu(dy).$$

Hence we obtain (A.7). Since

$$P_x(dy) = \frac{e^{\psi_0(y) - c_\lambda(x,y)/\epsilon}}{\int e^{\psi_0(y') - c_\lambda(x,y')/\epsilon} \nu_0(dy')} \nu_0(dy),$$

(A.7) simplifies to

$$\dot{\phi}(x) + \int \dot{\psi}(y) P_x(dy) = - \int p_x(y) h_\nu(dy), \quad (\text{A.8})$$

where $p_x(y) = dP_x/d\mu_0(y)$. Similarly, differentiating the second component yields

$$\dot{\psi}(y) + \int \dot{\phi}(x) Q_y(dx) = - \int q_y(x) h_\mu(dx), \quad (\text{A.9})$$

with $Q_y(dx) = \pi_0(dx|y)$ and $q_y(x) = dQ_y/d\mu_0(x)$. Equations (A.8)-(A.9) are the linearised system in the statement of Theorem 1. In operator form, using $E\varphi(x) = \int \varphi(y) P_x(dy)$ and $E^*\varphi(y) = \int \varphi(x) Q_y(dx)$, we rewrite them as:

$$\dot{\phi} + E\dot{\psi} = A, \quad \dot{\psi} + E^*\dot{\phi} = B, \quad (\text{A.10})$$

where $A(x) = - \int p_x(y) h_\nu(dy)$ and $B(y) = - \int q_y(x) h_\mu(dx)$. Substituting $\dot{\phi} = A - E\dot{\psi}$ in $\dot{\psi} + E^*\dot{\phi} = B$ and rearranging the terms gives $(I - E^*E)\dot{\psi} = B - E^*A$. Notice that A and B are mean-zero:

$$\int A(x) \mu_0(dx) = - \iint p_x(y) h_\nu(dy) \mu_0(dx) = - \int \left[\int p_x(y) \mu_0(dx) \right] h_\nu(dy) = - \int h_\nu(dy) = 0,$$

because $\int p_x(y) \mu_0(dx) = 1$ by the marginal property of the self-coupling π_0 . The same argument yields $\int B(y) \mu_0(dy) = 0$. Hence $A, B \in L_0^\infty(\mu_0)$. Since, by Lemma 2, E^* leaves $L_0^\infty(\mu_0)$ invariant, it follows that $E^*A \in L_0^\infty(\mu_0)$, and therefore $B - E^*A \in L_0^\infty(\mu_0)$. Still, Lemma 2 ensures that $I - E^*E$ is invertible on $L_0^\infty(\mu_0)$, so the Neumann series implies

$$\|(I - E^*E)^{-1}\|_{L_0^\infty(\mu_0)} \leq \sum_{k=0}^{\infty} \|E^*E\|_{L_0^\infty(\mu_0)}^k = \frac{1}{1 - \|E^*E\|_{L_0^\infty(\mu_0)}},$$

and using Lemma 2, we obtain

$$\|(I - E^*E)^{-1}\|_{L_0^\infty(\mu_0)} \leq \frac{1}{1 - (1 - m)^2}.$$

Since $A, B \in L^\infty$ whenever $\|h_\mu\|_{\text{TV}}, \|h_\nu\|_{\text{TV}} \leq 2$, we obtain $\dot{\psi} \in L^\infty$. Then $\dot{\phi} = A - E\dot{\psi} \in$

L^∞ because E is bounded on L^∞ . The uniform bound follows. That concludes the proof. $\square$

To make this abstract result operational, we need to provide an expression for the derivative relevant for the influence function and give the representation of Γ_{μ_0} . To this end, we need the derivative of the map $\mu \mapsto \psi_{\mu, \mu_\theta}$. Crucially, the second marginal μ_θ is held fixed when differentiating with respect to μ , so we set $h_\nu = 0$ in the linearised system (A.4)–(A.5). For point-mass perturbations such as $h_\mu = \delta_z - \mu_0$, the system has solution $(\dot{\phi}_z, \dot{\psi}_z)$. The second component gives the derivative of the second potential with respect to the first marginal:

$$\dot{\psi}_z(y) := D_\mu \psi_{\mu, \mu_{\theta_0}}[\delta_z - \mu_0](y) \Big|_{\mu=\mu_0},$$

which is the quantity required in the definition of Γ_{μ_0} . Notice that the signed measure $h = \delta_z - \mu$ has zero total mass:

$$h(\mathcal{X}) = \delta_z - \mu(\mathcal{X}) = 1 - 1 = 0,$$

and finite total variation norm $\|h\|_{\text{TV}} \leq \|\delta_z\|_{\text{TV}} + \|\mu\|_{\text{TV}} = 2$. Thus $h \in \mathcal{M}_0(\mathcal{X})$ and is a legitimate direction for the Fréchet derivative. Moreover, recall that Γ_{μ_0} is defined as the Fréchet derivative of

$$\mu \mapsto \int \psi_{\mu, \mu_{\theta_0}}(y) s_{\theta_0}(y) \mu_{\theta_0}(dy),$$

with respect to the first marginal μ , while the second marginal μ_{θ_0} is held fixed. The dominated convergence theorem can be applied to the derivative of the integrand $y \mapsto \psi_{\mu, \mu_{\theta_0}}(y) s_{\theta_0}(y)$. Indeed, Step 2 shows that $\|\dot{\psi}_z\|_{L^\infty} < \infty$, which combined with the score integrability $\int \|s_{\theta_0}\| d\mu_{\theta_0} < \infty$ furnishes an integrable dominating function. Hence, differentiating under the integral is justified and yields:

$$\Gamma_{\mu_0}(\delta_z - \mu_0) = \int D_\mu \psi_{\mu, \mu_{\theta_0}}[\delta_z - \mu_0](y) s_{\theta_0}(y) \mu_{\theta_0}(dy) = \int \dot{\psi}_z(y) s_{\theta_0}(y) \mu_{\theta_0}(dy),$$

which is Eq. (7) of Section 5 in the paper.

Proof of Corollary 1. For $h_\mu = \delta_z - \mu_0$ and $h_\nu = 0$, define

$$\begin{aligned} B_z(y) &= - \int q_y(x) (\delta_z - \mu_0)(dx) = - \int q_y(x) d\delta_z - \int q_y(x) \mu_0(dx) \\ &= -q_y(z) + Q_y(\mathcal{X}) = -q_y(z) + 1 \end{aligned} \tag{A.11}$$

where $Q_y(\mathcal{X})$ follows from the fact that q_y is the conditional density. Moreover, since $q_y(x)$ is uniformly bounded (Lemma 2), B_z is uniformly bounded in $L^\infty(\mathcal{X})$ over all $z \in \mathcal{X}$. Moreover,

$\dot{\psi}_z = (I - E^*E)^{-1}B_z$. By Lemma 2,

$$\|(I - E^*E)^{-1}\|_{L_0^\infty(\mu_0)} \leq \frac{1}{1 - (1 - m)^2},$$

and therefore

$$\|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} \leq \frac{\|B_z\|_{L^\infty(\mathcal{X})}}{1 - (1 - m)^2}.$$

Since $\sup_z \|B_z\|_{L^\infty(\mathcal{X})} < \infty$, it follows that $\sup_{z \in \mathcal{X}} \|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} < \infty$. Using the representation (A.3), we obtain

$$\sup_{z \in \mathcal{X}} \|\Gamma_{\mu_0}(\delta_z - \mu_0)\| \leq \sup_{z \in \mathcal{X}} \|\dot{\psi}_z\|_{L^\infty(\mathcal{X})} \int \|s_{\theta_0}(y)\| \mu_{\theta_0}(dy) < \infty.$$

The expression of the influence function follows from (A.3) together with

$$\text{IF}(z; T, \mu_0) = M(\theta_0)^{-1} \Gamma_{\mu_0}(\delta_z - \mu_0).$$

Therefore $\sup_{z \in \mathcal{X}} \|\text{IF}(z; T, \mu_0)\| < \infty$, so the E-ROBOT M-estimator is B -robust.

Remark 6. For $Z \sim \mu_0$, we have

$$\begin{aligned} \mathbb{E}_{\mu_0}[\text{IF}(Z; T, \mu_0)] &= M(\theta_0)^{-1} \int \Gamma_{\mu_0}(\delta_z - \mu_0) \mu_0(dz) \\ &= M(\theta_0)^{-1} \int \left[\int \dot{\psi}_z(y) s_{\theta_0}(y) \mu_0(dy) \right] \mu_0(dz) \\ &= M(\theta_0)^{-1} \int s_{\theta_0}(y) \left[\int \dot{\psi}_z(y) \mu_0(dz) \right] \mu_0(dy), \end{aligned}$$

where the last equality follows from Fubini–Tonelli theorem, applicable because $\dot{\psi}_z(y)$ is uniformly bounded and s_{θ_0} is μ_0 -integrable. From (A.11) we have $B_z(y) = 1 - q_y(z)$. Thus, integrating over z gives

$$\int B_z(y) \mu_0(dz) = \int (1 - q_y(z)) \mu_0(dz) = 1 - 1 = 0.$$

Recall from the linear-response system that $\dot{\psi}_z = (I - E^*E)^{-1}B_z$, therefore

$$\int \dot{\psi}_z(y) \mu_0(dz) = (I - E^*E)^{-1} \left[\int B_z(y) \mu_0(dz) \right] = (I - E^*E)^{-1}(0) = 0.$$

Substituting into the previous expression yields

$$\mathbb{E}_{\mu_0}[\text{IF}(Z; T, \mu_0)] = M(\theta_0)^{-1} \int s_{\theta_0}(y) \cdot 0 \mu_0(dy) = 0.$$

Thus the influence function has zero expected value under the true model, exactly as in classical M -estimation theory; see e.g. Hampel et al. (1986, p.101, Eq. (2.3.4)–(2.3.5)).

A.4 Proofs for Asymptotic Normality

Corollary 1 shows that the influence operator Γ_{μ_0} is bounded on point-mass contaminations, i.e., that its representer $\gamma(z) := \Gamma_{\mu_0}(\delta_z - \mu_0)$ satisfies $\sup_z \|\gamma(z)\| < \infty$. Boundedness of γ alone, however, is not sufficient to obtain the limiting distribution of $\widehat{\theta}_n$: the empirical measure μ_n converges to μ_0 only weakly, never in total variation ($\|\mu_n - \mu_0\|_{\text{TV}} = 2$ for every n), so a derivative that is only guaranteed to behave well under total-variation-small perturbations does not automatically control the limit of $\Psi_n(\theta_0) = \Psi(\mu_n, \theta_0)$. In this section, we close this gap.

First, we show that γ is not merely bounded but Lipschitz, which upgrades Γ_{μ_0} to a continuous linear functional with respect to the bounded-Lipschitz norm. Second, we establish Hadamard differentiability of the Schrödinger map in the bounded-Lipschitz topology and show that Γ_{μ_0} is the corresponding Hadamard derivative of $\mu \mapsto \Psi(\mu, \theta_0)$. This allows us to invoke the functional delta method and prove asymptotic normality.

To proceed with the proof, we recall that for a finite signed Borel measure h on $\mathcal{X}$ with $h(\mathcal{X}) = 0$, define the bounded-Lipschitz norm

$$\|h\|_{\text{BL}} := \sup \left\{ \left| \int f dh \right| : f \in \text{BL}_1(\mathcal{X}) \right\},$$

where $\text{BL}_1(\mathcal{X})$ denotes the unit ball of the space of bounded 1-Lipschitz functions on $\mathcal{X}$:

$$\text{BL}_1(\mathcal{X}) := \{f : \mathcal{X} \rightarrow \mathbb{R} : \|f\|_{\infty} \leq 1, |f(x) - f(x')| \leq \|x - x'\| \ \forall x, x' \in \mathcal{X}\}.$$

The $\|\cdot\|_{\text{BL}}$ metrises weak convergence on $\mathcal{P}(\mathcal{X})$, and $\|h\|_{\text{BL}} \leq \|h\|_{\text{TV}}$.

A.4.1 Auxiliary lemmas about Lipschitz conditions

Lemma 7. *Let μ, ν be probability measures on a Polish space $\mathcal{X}$, and let (ϕ, ψ) be the optimal Schrödinger potentials for the E-ROBOT cost c_{λ} with regularization $\epsilon > 0$, normalized by $\int \phi d\mu = 0$. For the self-coupling pair (μ_0, μ_0) , the optimal coupling is*

$$\pi_0(dx, dy) = e^{(\phi_0(x) + \psi_0(y) - c_{\lambda}(x, y))/\epsilon} \mu_0(dx) \mu_0(dy),$$

and the conditional densities are

$$p_x(y) = \frac{e^{\psi_0(y)/\epsilon} K(x, y)}{Z(x)}, \quad Z(x) = \int e^{\psi_0(y')/\epsilon} K(x, y') \mu_0(dy'),$$

$$q_y(x) = \frac{e^{\phi_0(x)/\epsilon} K(x, y)}{Z'(y)}, \quad Z'(y) = \int e^{\phi_0(x')/\epsilon} K(x', y) \mu_0(dx').$$

Then, there exists a constant $\kappa_1 = \kappa_1(\epsilon, \lambda)$, independent of $\mathcal{X}$ and the dimension, such that

$$\sup_{y \in \mathcal{X}} |q_y(x) - q_y(x')| \leq \kappa_1 |x - x'|, \quad \sup_{x \in \mathcal{X}} |p_x(y) - p_x(y')| \leq \kappa_1 |y - y'|,$$

for all $x, x', y, y' \in \mathcal{X}$.

Proof. We make use of the bounds in (A.1) and we specialize them to (μ_0, μ_0) , letting (ϕ_0, ψ_0) be the optimal potentials normalized by $\int \phi_0 d\mu_0 = 0$. From (A.1) we know they are bounded and Lipschitz with constants C_0, κ_0 depending only on ϵ, λ . We show that both $p_x(y)$ and $q_y(x)$ are Lipschitz in the appropriate variable with a uniform constant. First, note that $Z(x)$ is bounded away from zero and infinity:

$$m_K e^{-C_0/\epsilon} \leq Z(x) \leq e^{C_0/\epsilon}.$$

Also, by the Lipschitz property of ψ_0 and K , the map $x \mapsto e^{\psi_0(y)/\epsilon} K(x, y)$ is Lipschitz in x uniformly in y : for any x, x', y ,

$$|e^{\psi_0(y)/\epsilon} K(x, y) - e^{\psi_0(y)/\epsilon} K(x', y)| \leq e^{C_0/\epsilon} \frac{1}{\epsilon} |x - x'|.$$

Furthermore, $Z(x)$ is Lipschitz:

$$|Z(x) - Z(x')| \leq \int e^{\psi_0(y')/\epsilon} |K(x, y') - K(x', y')| \mu_0(dy') \leq e^{C_0/\epsilon} \frac{1}{\epsilon} |x - x'|.$$

Therefore, using the quotient bound

$$\left| \frac{a}{b} - \frac{a'}{b'} \right| \leq \frac{|a - a'|}{b} + \frac{a' |b - b'|}{bb'}$$

with $b, b' \geq m_K e^{-C_0/\epsilon}$, we obtain

$$|p_x(y) - p_{x'}(y)| \leq \kappa_1 |x - x'|$$

for a constant

$$\kappa_1 = \frac{e^{2C_0/\epsilon}}{\epsilon m_K} + \frac{e^{4C_0/\epsilon}}{\epsilon m_K^2}.$$

The same argument applied to $q_y(x)$ (now using the Lipschitz property of ϕ_0 and the same bounds for $Z'(y)$) yields

$$|q_y(x) - q_y(x')| \leq \kappa_1 |x - x'|.$$

The symmetric inequalities in y for $p_x(y)$ and in x for $q_y(x)$ follow identically. This completes the proof. $\square$

A.4.2 Hadamard differentiability in BL norm

The following result provides the Hadamard differentiability bounded-Lipschitz norm required for the functional delta method. Following the proof of Theorem 1, the subsequent proof relies on an implicit function theorem argument that yields Fréchet differentiability in the bounded-Lipschitz (BL) norm, and hence Hadamard differentiability, since Fréchet implies Hadamard on

any normed space.

Proposition 8. *Let $(\mu_t)_{t>0} \subset \mathcal{P}(\mathcal{X})$ be such that $\mu_t = \mu_0 + t h_t$ for some $h_t \in \mathcal{M}_0(\mathcal{X})$ with $h_t \rightarrow h$ in $\|\cdot\|_{\text{BL}}$ as $t \rightarrow 0$, for some h in the $\|\cdot\|_{\text{BL}}$ -closure of $\mathcal{M}_0(\mathcal{X})$. Then*

$$t^{-1}[\Psi(\mu_t, \theta_0) - \Psi(\mu_0, \theta_0)] \rightarrow \Gamma_{\mu_0}(h) \quad \text{as } t \rightarrow 0.$$

Proof. The proof is a direct application of the Banach-space implicit function theorem in the bounded-Lipschitz topology.

Step 1. Consider the Schrödinger fixed-point equations

$$\phi(x) = -\epsilon \log \int e^{\psi(y)/\epsilon} K(x, y) d\mu_0(y), \quad \psi(y) = -\epsilon \log \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x),$$

with $K(x, y) = e^{-c\lambda(x, y)/\epsilon}$. Write the right-hand side as the operator $\mathcal{F}(\phi, \psi; \mu, \mu_0)$ defined in (A.6), but now with the second marginal fixed at μ_0 , namely

$$\mathcal{F}(\phi, \psi; \mu, \mu_0) = \begin{pmatrix} -\epsilon \log \int e^{\psi(y)/\epsilon} K(\cdot, y) d\mu_0(y) \\ -\epsilon \log \int e^{\phi(x)/\epsilon} K(x, \cdot) d\mu(x) \end{pmatrix}.$$

Here the first component does not depend on μ , while the second component is affine in μ . By Lemma 7, the potentials are uniformly bounded and uniformly Lipschitz with constants depending only on ϵ, λ . The differentiability of $\mathcal{F}$ with respect to the potentials (ϕ, ψ) has already been mentioned in the proof of Theorem 1 and is detailed in Appendix B; in particular, for fixed measures, the map $(\phi, \psi) \mapsto \mathcal{F}(\phi, \psi; \mu, \mu_0)$ is continuously Fréchet differentiable in the uniform norm. It remains to verify the differentiability with respect to the measure argument μ .

To see this, observe that the second component of $\mathcal{F}$ is

$$\mathcal{F}_2(\phi, \psi; \mu, \mu_0)(y) = -\epsilon \log \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x).$$

For fixed (ϕ, ψ) , the perturbation $\mu + th$ gives

$$\int e^{\phi(x)/\epsilon} K(x, y) d(\mu + th)(x) = \int e^{\phi(x)/\epsilon} K(x, y) d\mu(x) + t \int e^{\phi(x)/\epsilon} K(x, y) h(dx).$$

Hence, by the chain rule,

$$\frac{d}{dt} \left[-\epsilon \log \int e^{\phi(x)/\epsilon} K(x, y) d(\mu + th)(x) \right]_{t=0} = -\epsilon \frac{\int e^{\phi(x)/\epsilon} K(x, y) h(dx)}{\int e^{\phi(x)/\epsilon} K(x, y) d\mu(x)}.$$

The first component of $\mathcal{F}$ does not depend on μ , so its derivative with respect to μ is zero. This

yields that the derivative with respect to μ in a direction h is

$$h \mapsto \begin{pmatrix} 0 \\ -\epsilon \frac{\int e^{\phi(x)/\epsilon} K(x, \cdot) h(dx)}{\int e^{\phi(x)/\epsilon} K(x, \cdot) d\mu(x)} \end{pmatrix}.$$

By Lemma 7, the integrand $x \mapsto e^{\phi(x)/\epsilon} K(x, y)$ is uniformly bounded and uniformly Lipschitz, so its bounded-Lipschitz norm is bounded by a constant C_K independent of y and of (ϕ, ψ) . Moreover, the denominator is uniformly bounded below by $e^{-C_0/\epsilon} e^{-2\lambda/\epsilon} > 0$. Therefore, for every h ,

$$\left| -\epsilon \frac{\int e^{\phi(x)/\epsilon} K(x, y) h(dx)}{\int e^{\phi(x)/\epsilon} K(x, y) d\mu(x)} \right| \leq \epsilon \frac{C_K}{e^{-(C_0+2\lambda)/\epsilon}} \|h\|_{\text{BL}},$$

uniformly in y , which proves the claim.

At the base point $(\phi_0, \psi_0, \mu_0, \mu_0)$ with $\phi_0 = \phi_{\mu_0, \mu_0}$ and $\psi_0 = \psi_{\mu_0, \mu_0}$, the derivative with respect to (ϕ, ψ) is the block operator

$$\begin{pmatrix} I & E \\ E^* & I \end{pmatrix},$$

where E and E^* are the conditional expectation operators introduced before Theorem 1; see main paper, Section 5. By Lemma 2, the linearized system

$$u + Ev = r, \quad v + E^*u = s$$

is uniquely solvable for every right-hand side (r, s) satisfying $\int r d\mu_0 = \int s d\mu_0$, with solution $(u, v) \in \mathcal{B}_0$. Hence the derivative is a linear homeomorphism from $\mathcal{B}_0$ onto the closed subspace

$$\left\{ (r, s) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{X}) : \int r d\mu_0 = \int s d\mu_0 \right\},$$

and the Banach-space implicit function theorem (see e.g. [1], Ch. X, p.270) applies. Thus, there exists a neighborhood of μ_0 in the bounded-Lipschitz topology such that the solution map $\mu \mapsto (\phi_{\mu, \mu_0}, \psi_{\mu, \mu_0})$ is Fréchet differentiable at μ_0 . Its derivative in a direction h is the solution $(\dot{\phi}_h, \dot{\psi}_h)$ of the linearised system $\dot{\phi}_h + E\dot{\psi}_h = 0$, and $\dot{\psi}_h + E^*\dot{\phi}_h = B_h$. This is exactly the linearised system from Theorem 1 specialised to $h_\mu = h$, $h_\nu = 0$. Moreover, because the derivative is a bounded linear operator, the solution map is Lipschitz continuous near μ_0 :

$$\|\phi_{\mu, \mu_0} - \phi_{\mu_0, \mu_0}\|_\infty \vee \|\psi_{\mu, \mu_0} - \psi_{\mu_0, \mu_0}\|_\infty \leq C \|\mu - \mu_0\|_{\text{BL}}$$

for some $C < \infty$. In particular, for the path $\mu_t \rightarrow \mu_0$ in BL we have

$$\|\phi_t - \phi_0\|_\infty \vee \|\psi_t - \psi_0\|_\infty \rightarrow 0.$$

Step 2. By Lemma 1,

$$\Psi(\mu, \theta_0) = \int (\psi_{\mu, \mu_0}(y) - u_{\mu_0}(y)) s_{\theta_0}(y) \mu_{\theta_0}(dy).$$

The term involving u_{μ_0} is independent of μ . Since $\mu \mapsto \psi_{\mu, \mu_0}$ is Fréchet differentiable in BL and the integration against $s_{\theta_0} \mu_{\theta_0}$ is a bounded linear functional, the composition is Fréchet differentiable in BL with derivative

$$\Gamma_{\mu_0}(h) = \int \dot{\psi}_h(y) s_{\theta_0}(y) \mu_{\theta_0}(dy).$$

This is the same operator Γ_{μ_0} as defined in (A.3). Fréchet differentiability implies Hadamard differentiability, since we are working in the normed space given by the $\|\cdot\|_{\text{BL}}$ -closure of $\mathcal{M}_0(\mathcal{X})$, and the derivative is identical. Therefore,

$$t^{-1}[\Psi(\mu_t, \theta_0) - \Psi(\mu_0, \theta_0)] \rightarrow \Gamma_{\mu_0}(h)$$

whenever $\mu_t = \mu_0 + th_t$ with $h_t \rightarrow h$ in BL. This proves the proposition. $\square$

Remark 9. In Theorem 1 we establish Fréchet differentiability in the TV norm, which gives the uniform bound over point masses used for B-robustness. The BL Fréchet differentiability shown here is the appropriate tool for the asymptotic distribution, because the empirical process converges in BL but not in TV. Thanks to this, we can apply the functional delta method via Hadamard differentiability in BL norm. Note that the derivative operator Γ_{μ_0} is the same in both cases.

A.4.3 Lipschitz continuity of the influence function

Similarly to Corollary 1 to Theorem 1 for the TV norm, we state the following corollary to Proposition 1 for the BL norm:

Corollary 10. Under the assumptions of Proposition 1, the representer

$$\gamma(z) := \Gamma_{\mu_0}(\delta_z - \mu_0) = M(\theta_0) \text{IF}(z; T, \mu_0)$$

is Lipschitz:

$$\|\gamma(z) - \gamma(z')\| \leq \kappa_\gamma \|z - z'\|,$$

where

$$\kappa_\gamma := \frac{\kappa_1}{1 - (1 - m)^2} \|\nabla_{\theta_0} \mu_{\theta_0}\|_{\text{TV}},$$

κ_1 is the constant of Lemma 7 and m is the Doeblin constant in (A.3). Consequently, for every $h \in \mathcal{M}_0(\mathcal{X})$ with $\|h\|_{\text{BL}} < \infty$, the operator Γ_{μ_0} extends continuously to the bounded-Lipschitz

topology and satisfies

$$|\Gamma_{\mu_0}(h)| \leq \left(\sup_z \|\gamma(z)\| \vee \kappa_\gamma \right) \|h\|_{\text{BL}} < \infty.$$

Proof. From the proof of Corollary 1,

$$\gamma(z) = \Gamma_{\mu_0}(\delta_z - \mu_0) = \int \dot{\psi}_z(y) s_{\theta_0}(y) \mu_{\theta_0}(dy),$$

with

$$\dot{\psi}_z = (I - E^* E)^{-1} B_z,$$

where B_z is defined in (A.11). Moreover,

$$B_z(y) = - \int q_y(x) (\delta_z - \mu_0)(dx) = -q_y(z) + \int q_y(x) \mu_0(dx),$$

using $\int q_y(x) \mu_0(dx) = 1$, since Q_y is a probability kernel.

By Lemma 7,

$$\|B_z - B_{z'}\|_{L^\infty(\mathcal{X})} = \sup_{y \in \mathcal{X}} |q_y(z) - q_y(z')| \leq \kappa_1 \|z - z'\|.$$

Using the resolvent bound from Lemma 2,

$$\|\dot{\psi}_z - \dot{\psi}_{z'}\|_{L^\infty(\mathcal{X})} = \|(I - E^* E)^{-1}(B_z - B_{z'})\|_{L^\infty(\mathcal{X})} \leq \frac{\kappa_1}{1 - (1 - m)^2} \|z - z'\|.$$

Consequently,

$$\|\gamma(z) - \gamma(z')\| \leq \|\dot{\psi}_z - \dot{\psi}_{z'}\|_{L^\infty(\mathcal{X})} \int \|s_{\theta_0}(y)\| \mu_{\theta_0}(dy),$$

and therefore

$$\|\gamma(z) - \gamma(z')\| \leq \frac{\kappa_1}{1 - (1 - m)^2} \|\nabla_\theta \mu_{\theta_0}\|_{\text{TV}} \|z - z'\| = \kappa_\gamma \|z - z'\|.$$

For the extension claim, note that Corollary 1 implies

$$\sup_z \|\gamma(z)\| < \infty,$$

and we have just shown that γ is Lipschitz with constant $\kappa_\gamma < \infty$. Hence each component of

$$\frac{\gamma}{\sup_z \|\gamma(z)\| \vee \kappa_\gamma}$$

belongs to $\text{BL}_1(\mathcal{X})$. By the definition of the bounded-Lipschitz norm, for every $h \in \mathcal{M}_0(\mathcal{X})$,

$$|\Gamma_{\mu_0}(h)| = \left| \int \gamma dh \right| \leq \left(\sup_z \|\gamma(z)\| \vee \kappa_\gamma \right) \|h\|_{\text{BL}}.$$

This completes the proof. $\square$

A.4.4 Proof of Theorem 2

We denote by $\ell^\infty(\text{BL}_1(\mathcal{X}))$ the Banach space of all bounded real-valued mappings on $\text{BL}_1(\mathcal{X})$, equipped with the supremum norm

$$\|Z\|_{\ell^\infty(\text{BL}_1)} = \sup_{f \in \text{BL}_1(\mathcal{X})} |Z(f)|.$$

The empirical process $\mathbb{G}_n(f) = \sqrt{n}(\mu_n - \mu_0)(f) = \sqrt{n} \int f d(\mu_n - \mu_0)$ is viewed as a random element of $\ell^\infty(\text{BL}_1(\mathcal{X}))$; see e.g. [van der Vaart and Wellner \(1996, Sections 2.1\)](#).

To pass from Proposition 8 to a distributional limit for $\Psi_n(\theta_0)$, we need $\sqrt{n}(\mu_n - \mu_0)$ to converge weakly with respect to $\|\cdot\|_{\text{BL}}$ to a μ_0 -Brownian bridge. This is automatic when $\mathcal{X}$ is compact; on $\mathcal{X} = \mathbb{R}^d$ it holds under a mild tail condition. Namely, either (a) $\mathcal{X}$ is compact; or (b) $\mathcal{X} = \mathbb{R}^d$ and μ_0 satisfies a condition sufficient for $\text{BL}_1(\mathcal{X})$ to be μ_0 -Donsker. As far as (a) is concerned, when $\mathcal{X}$ is compact, $\text{BL}_1(\mathcal{X})$ is totally bounded in the sup norm, hence μ_0 -Donsker for every $\mu_0 \in \mathcal{P}(\mathcal{X})$ ([van der Vaart and Wellner, 1996, Ch 2.5](#)). As far as (b) is concerned, for example, it is enough that

$$\int \sqrt{1 + \|x\|^2} d\mu_0(x) < \infty$$

(Dudley's sufficient condition), which is satisfied by some heavy-tailed distributions, including the t_2 -distribution considered in the numerical examples of Section 6 in the main paper. Weaker tail conditions also suffice; for instance, the condition is automatically satisfied whenever μ_0 has finite second moment. Under this condition, the functional central limit theorem ([van der Vaart and Wellner, 1996, Sections 2.1, 2.8.2, and 2.8.3](#)) yields

$$\sqrt{n}(\mu_n - \mu_0) \xrightarrow{d} \mathbb{G}_{\mu_0} \quad \text{in } \ell^\infty(\text{BL}_1(\mathcal{X})),$$

where $\mathbb{G}_{\mu_0}$ is a tight, mean-zero μ_0 -Brownian bridge. Thanks to this result, we can proceed with the proof of Theorem 2 by invoking the functional delta method.

Proof. By Proposition 8 and the functional delta method ([van der Vaart and Wellner, 1996, Theorem 3.9.4](#)), the Hadamard differentiability of $\Psi(\cdot, \theta_0)$ at μ_0 together with the weak convergence $\mathbb{G}_n = \sqrt{n}(\mu_n - \mu_0) \Rightarrow \mathbb{G}_{\mu_0}$ imply

$$\sqrt{n} \Psi_n(\theta_0) = \sqrt{n}(\Psi(\mu_n, \theta_0) - \Psi(\mu_0, \theta_0)) \Rightarrow \Gamma_{\mu_0}(\mathbb{G}_{\mu_0}).$$

By Corollary 10, Γ_{μ_0} is a bounded linear functional with representer $\gamma(z) = \Gamma_{\mu_0}(\delta_z - \mu_0)$. Hence we write

$$\Gamma_{\mu_0}(\mathbb{G}_{\mu_0}) = \int \gamma d\mathbb{G}_{\mu_0} = \mathbb{G}_{\mu_0}(\gamma),$$

which is a centered Gaussian random vector. Recall (see e.g. [van der Vaart and Wellner \(1996, p.82\)](#)) that the covariance structure is given by:

$$\text{Cov}(\mathbb{G}_{\mu_0}(f), \mathbb{G}_{\mu_0}(g)) = \int fg d\mu_0 - \left(\int f d\mu_0 \right) \left(\int g d\mu_0 \right), \quad (\text{A.12})$$

for all bounded measurable functions f, g . Since $\Gamma_{\mu_0}(\mathbb{G}_{\mu_0}) = \mathbb{G}_{\mu_0}(\gamma)$, in our setting the covariance is

$$\Sigma_0 := \text{Cov}(\Gamma_{\mu_0}(\mathbb{G}_{\mu_0})) = \text{Cov}(\mathbb{G}_{\mu_0}(\gamma)).$$

Applying (A.12) coordinatewise to the vector-valued function γ , we obtain

$$\Sigma_0 = \int \gamma(z)\gamma(z)^\top \mu_0(dz) - \left(\int \gamma(z) \mu_0(dz) \right) \left(\int \gamma(z) \mu_0(dz) \right)^\top.$$

Equivalently, if $Z \sim \mu_0$,

$$\Sigma_0 = \text{Cov}_{\mu_0}(\gamma(Z)).$$

For a point-mass perturbation at z , we have $\Gamma_{\mu_0}(\delta_z - \mu_0) = \gamma(z) - \int \gamma(z) \mu_0(dz)$. Therefore, if $Z \sim \mu_0$,

$$\Gamma_{\mu_0}(\delta_Z - \mu_0) = \gamma(Z) - \mathbb{E}_{\mu_0}[\gamma(Z)],$$

and consequently

$$\text{Cov}(\Gamma_{\mu_0}(\delta_Z - \mu_0)) = \text{Cov}_{\mu_0}(\gamma(Z)) = \Sigma_0.$$

Since γ is bounded (Corollary 1), the covariance matrix Σ_0 is finite. Thus

$$\sqrt{n} \Psi_n(\theta_0) \Rightarrow N(0, \Sigma_0).$$

Thus the estimating equation satisfies a multivariate central limit theorem with asymptotic covariance matrix Σ_0 . Consistency of $\hat{\theta}_n$ follows from the general minimum-distance argument recalled in Section 4 of the main text (adapting Theorems 2.1–2.2 of [Bernton et al., 2019](#)). From this point on, we proceed as in standard M-estimation theory; see e.g. [Hampel et al. \(1986\)](#); [La Vecchia \(2016\)](#). The usual Taylor expansion of Ψ_n around θ_0 yields

$$0 = \Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + \nabla_{\theta} \Psi_n(\theta_n^*)(\hat{\theta}_n - \theta_0)$$

for some θ_n^* between $\hat{\theta}_n$ and θ_0 . Consistency of $\hat{\theta}_n$ together with the assumed uniform convergence

$$\nabla_{\theta} \Psi_n(\theta) \rightarrow M(\theta_0)$$

implies

$$\nabla_{\theta} \Psi_n(\theta_n^*) \rightarrow M(\theta_0)$$

in probability. Since $M(\theta_0)$ is nonsingular,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\nabla_{\theta} \Psi_n(\theta_n^*)^{-1} \sqrt{n} \Psi_n(\theta_0) \Rightarrow -M(\theta_0)^{-1} N(0, \Sigma_0),$$

and therefore, by Slutsky's theorem,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, M(\theta_0)^{-1} \Sigma_0 M(\theta_0)^{-1}).$$

Thus

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, V(\theta_0)),$$

where

$$V(\theta_0) = M(\theta_0)^{-1} \Sigma_0 M(\theta_0)^{-1}.$$

This is the claim of Theorem 2. □

A.5 Numerical implementation aspects

The base E-ROBOT Sinkhorn routine (Algorithm 1) computes the entropic optimal transport plan and cost for two marginals using the truncated cost $c_{\lambda}(x, y) = \min\{\|x - y\|, 2\lambda\}$. To compute the debiased divergence $\overline{W}_{\epsilon, \lambda}(\mu, \nu)$, the entropic cost $W_{\epsilon, \lambda}$ must be evaluated for three pairs of marginals: (μ, ν) , (μ, μ) , and (ν, ν) . For further details about $\overline{W}_{\epsilon, \lambda}$, we refer to [La Vecchia and Liu \(2026\)](#). Here for the sake of completeness, we recall the key steps of the Sinkhorn procedure.

Algorithm 2 implements the E-ROBOT Sinkhorn routine for each pair and combining the resulting costs.

Equipped with the debiased divergence, then we can proceed to the optimization of the loss, obtaining the E-ROBOT M-estimator $\hat{\theta}_n$. We provide a pseudo code to explain how to implement the method. Algorithm 3 wraps the divergence computation into a numerical optimisation routine, for instance via the `optim` function in R with the `Nelder-Mead` method.

Algorithm 1 E-ROBOT Sinkhorn Algorithm

```
Input:
   $1_n = (1, \dots, 1)^\top$ 
  Source marginal:  $\mu \in \mathbb{R}_+^n$ , where  $\mu^\top 1_n = 1$ 
  Target marginal:  $\nu \in \mathbb{R}_+^n$ , where  $\nu^\top 1_n = 1$ 
  Regularization parameter:  $\varepsilon > 0$ 
  Robustness parameter:  $\lambda > 0$ 
Output:
  Approximate optimal transport plan:  $\pi^{(t)}$ 
  Final scaling vectors:  $u^{(t)}, v^{(t)}$ 
  Entropic optimal transport cost:  $W_{\varepsilon, \lambda}(\mu, \nu)$ 
% Precompute the Gibbs kernel matrix  $K$  and cost matrix  $C_\lambda$ 
for  $i = 1$  to  $n$  do
  for  $j = 1$  to  $n$  do
     $c_\lambda(i, j) \leftarrow \min(\|x_i - y_j\|, 2\lambda)$ 
     $K(i, j) \leftarrow \exp(-c_\lambda(i, j)/\varepsilon)$ 
  end for
end for
% Initialize scaling vectors
 $u^{(0)} \leftarrow 1_n$ 
% Main Sinkhorn iteration loop
for  $t = 0, 1, 2, \dots$  do
   $v^{(t)} \leftarrow \nu / (K^\top u^{(t)})$ 
   $u^{(t+1)} \leftarrow \mu / (K v^{(t)})$ 
end for
% Form the approximate optimal transport plan
 $\pi^{(t)} \leftarrow \text{diag}(u^{(t)}) K \text{diag}(v^{(t)})$ 
% Compute the entropic optimal transport cost
 $C_{\text{linear}} \leftarrow \sum_{i=1}^n \sum_{j=1}^n c_\lambda(i, j) \pi^{(t)}(i, j)$ 
 $H \leftarrow \sum_{i=1}^n \sum_{j=1}^n \pi^{(t)}(i, j) \log\left(\frac{\pi^{(t)}(i, j)}{\mu^{(i)} \nu^{(j)}}\right)$ 
 $W_{\varepsilon, \lambda}(\mu, \nu) \leftarrow C_{\text{linear}} + \varepsilon H$ 
return  $\pi^{(t)}, u^{(t)}, v^{(t)}, W_{\varepsilon, \lambda}(\mu, \nu)$ 
```

Algorithm 2 E-ROBOT Debiased Divergence

Input:
Source marginal: $\mu \in \mathbb{R}_+^n$
Target marginal: $\nu \in \mathbb{R}_+^n$
Regularization: $\varepsilon > 0$, robustness: $\lambda > 0$
Output: Debiased divergence $\bar{W}_{\varepsilon, \lambda}(\mu, \nu)$
procedure EROBOT-Debiased($\mu, \nu, \varepsilon, \lambda$)
 % Compute the three required Sinkhorn plans and entropic costs
 $(\pi_{\mu\nu}, u_{\mu\nu}, v_{\mu\nu}, W_{\mu\nu}) \leftarrow \text{EROBOT-Sinkhorn}(\mu, \nu, \varepsilon, \lambda)$
 $(\pi_{\mu\mu}, u_{\mu\mu}, v_{\mu\mu}, W_{\mu\mu}) \leftarrow \text{EROBOT-Sinkhorn}(\mu, \mu, \varepsilon, \lambda)$
 $(\pi_{\nu\nu}, u_{\nu\nu}, v_{\nu\nu}, W_{\nu\nu}) \leftarrow \text{EROBOT-Sinkhorn}(\nu, \nu, \varepsilon, \lambda)$
 % Form the debiased divergence
 $\bar{W} \leftarrow W_{\mu\nu} - 0.5 \cdot (W_{\mu\mu} + W_{\nu\nu})$
 return $\bar{W}$
end procedure

Algorithm 3 E-ROBOT M-Estimator via Numerical Optimisation

Input:
Empirical measure: $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$
Generative model indexed by θ : $\mu_\theta, \theta \in \Theta \subset \mathbb{R}^k$
Initial parameter: θ_{init}
Regularization: $\varepsilon > 0$, robustness: $\lambda > 0$
Number of samples from μ_θ for discretisation: m
Output: Minimizer $\hat{\theta}_n$
procedure EROBOT-M-Estimator($\mu_n, \theta_{\text{init}}, \varepsilon, \lambda, m$)
 % Define the loss function
 $\text{loss}(\theta) \leftarrow \text{EROBOT-Debiased}(\mu_n, \hat{\mu}_\theta, \varepsilon, \lambda)$
 where $\hat{\mu}_\theta$ is the empirical measure of m i.i.d. draws from μ_θ
 % Minimise the loss using a derivative-free optimiser
 $\hat{\theta}_n \leftarrow \text{Optimise}(\text{loss}, \theta_{\text{init}}, \text{method} = \text{"Nelder-Mead"})$
 return $\hat{\theta}_n$
end procedure

Appendix 1

We provide sketchy derivation of Eq. (3) in the main paper. This derivation is heuristic and is obtained in analogy to the derivation available in von Mises calculus for M-functionals; see e.g. [van der Vaart \(1998, Ch. 20, p.291\)](#). The main difference with the existing approach is that, for E-ROBOT M-estimators, the first-order von Mises expansion does not involve an integral of a fixed function against the (signed) measure h ; instead, the integrand depends on the base measure μ through the Schrödinger potentials. As a consequence, the usual observation-wise form of the influence function for M-estimators does not directly apply: we obtain a different form for the influence function. The justification of the derivation sketched here is, essentially, based on the Banach-space implicit function theorem and it is given in the proofs of Lemma 1, Lemma 2, and Theorem 1.

Let $\Psi(\mu, \theta) = \nabla_{\theta} \overline{W}_{\epsilon, \lambda}(\mu, \mu_{\theta})$ be the population estimating equation. Fisher consistency gives $\Psi(\mu_0, \theta_0) = 0$. For a contamination point $z \in \mathcal{X}$, define the measure

$$\mu_t = (1 - t)\mu_0 + t\delta_z, = \mu_0 + t(\delta_z - \mu_0), \quad \text{for } t \ll \frac{1}{2},$$

and let $\theta_t = T(\mu_t)$ be the minimiser of the loss $\overline{W}_{\epsilon, \lambda}(\mu_t, \mu_{\theta})$. Then θ_t satisfies the first-order condition

$$\Psi(\mu_t, \theta_t) = 0 \quad \text{for all } t.$$

Assuming θ_t is differentiable in t , we differentiate with respect to t at $t = 0$:

$$\left. \frac{d}{dt} \Psi(\mu_t, \theta_t) \right|_{t=0} = 0.$$

By the chain rule,

$$\left. \frac{\partial \Psi}{\partial \mu} \right|_{\mu_0} \left(\left. \frac{d\mu_t}{dt} \right|_{t=0} \right) + \left. \frac{\partial \Psi}{\partial \theta} \right|_{\mu_0, \theta_0} \left(\left. \frac{d\theta_t}{dt} \right|_{t=0} \right) = 0. \quad (.13)$$

Now, set

$$\left. \frac{d\mu_t}{dt} \right|_{t=0} = \delta_z - \mu_0,$$

and

$$\left. \frac{d\theta_t}{dt} \right|_{t=0} = \text{IF}(z; T, \mu_0)$$

by the definition of the influence function; see [Hampel et al. \(1986\)](#). Moreover, since $\Psi = \nabla_{\theta} \overline{W}_{\epsilon, \lambda}$,

$$\left. \frac{\partial \Psi}{\partial \theta} \right|_{\mu_0} = \nabla_{\theta\theta}^2 \overline{W}_{\epsilon, \lambda}(\mu_0, \mu_{\theta_0}) =: -M(\theta_0).$$

Substituting these into (.13) yields

$$\left. \frac{\partial \Psi}{\partial \mu} \right|_{\mu_0} (\delta_z - \mu_0) + M(\theta_0) \text{IF}(z; T, \mu_0) = 0.$$

Finally, assuming $M(\theta_0)$ is nonsingular, we solve for the influence function:

$$\text{IF}(z; T, \mu_0) = M(\theta_0)^{-1} \left. \frac{\partial \Psi}{\partial \mu} \right|_{\mu_0, \theta_0} (\delta_z - \mu_0),$$

which is formula (3) in the main text.

Appendix 2

To justify that $\mathcal{F}$ is continuously Fréchet differentiable, we consider the first component of $\mathcal{F}$:

$$\mathcal{F}_\nu(\psi)(x) := -\epsilon \log \int e^{(\psi(y) - c_\lambda(x, y))/\epsilon} d\nu(y).$$

For fixed $\psi_0 \in L^\infty(\mathcal{X})$, we compute the Fréchet derivative of $\mathcal{F}_\nu$ at ψ_0 in direction $h \in L^\infty(\mathcal{X})$. Define

$$G_x(t) := \int e^{(\psi_0(y) + th(y) - c_\lambda(x, y))/\epsilon} d\nu(y).$$

Then

$$\mathcal{F}_\nu(\psi_0 + th)(x) = -\epsilon \log G_x(t).$$

Since c_λ is bounded by 2λ , we have

$$e^{-(\|\psi_0\|_\infty + 2\lambda)/\epsilon} \leq G_x(0) \leq e^{\|\psi_0\|_\infty/\epsilon},$$

so $G_x(0)$ is uniformly bounded away from zero and infinity in x .

By the dominated convergence theorem,

$$G'_x(0) = \frac{1}{\epsilon} \int h(y) e^{(\psi_0(y) - c_\lambda(x, y))/\epsilon} d\nu(y).$$

Hence

$$D\mathcal{F}_\nu(\psi_0)x = -\epsilon \frac{G'_x(0)}{G_x(0)} = -\frac{\int h(y) e^{(\psi_0(y) - c_\lambda(x, y))/\epsilon} d\nu(y)}{\int e^{(\psi_0(y) - c_\lambda(x, y))/\epsilon} d\nu(y)}.$$

This is the candidate Fréchet derivative. To verify Fréchet differentiability, note that the maps

$$u \mapsto e^u, \quad u \mapsto \log u,$$

are twice continuously differentiable. Since the denominator is uniformly bounded away from

zero, a second-order Taylor expansion yields

$$\|\mathcal{F}_\nu(\psi_0 + h) - \mathcal{F}_\nu(\psi_0) - D\mathcal{F}_\nu(\psi_0)[h]\|_\infty = O(\|h\|_\infty^2),$$

as $\|h\|_\infty \rightarrow 0$. Therefore

$$\frac{\|\mathcal{F}_\nu(\psi_0 + h) - \mathcal{F}_\nu(\psi_0) - D\mathcal{F}_\nu(\psi_0)[h]\|_\infty}{\|h\|_\infty} \rightarrow 0,$$

which proves that $\mathcal{F}_\nu$ is Fréchet differentiable.

Next, for any $h \in L^\infty(\mathcal{X})$,

$$|D\mathcal{F}_\nu(\psi_0)x| \leq \frac{\|h\|_\infty \int e^{(\psi_0(y) - c_\lambda(x,y))/\epsilon} d\nu(y)}{\int e^{(\psi_0(y) - c_\lambda(x,y))/\epsilon} d\nu(y)} = \|h\|_\infty.$$

Thus $D\mathcal{F}_\nu(\psi_0)$ is a bounded linear operator on $L^\infty(\mathcal{X})$ and

$$\|D\mathcal{F}_\nu(\psi_0)\|_{\mathcal{L}(L^\infty)} = \sup_{\|h\|_\infty \leq 1} \|D\mathcal{F}_\nu(\psi_0)[h]\|_\infty \leq 1.$$

We now show that the map

$$\psi \mapsto D\mathcal{F}_\nu(\psi)$$

is continuous in operator norm.

Let

$$\psi_1, \psi_2 \in L^\infty(\mathcal{X})$$

with

$$\|\psi_1 - \psi_2\|_\infty \leq 1,$$

and let

$$\|h\|_\infty \leq 1.$$

Write

$$a = \int h(y) e^{(\psi_1(y) - c_\lambda(x,y))/\epsilon} d\nu(y), \quad b = \int e^{(\psi_1(y) - c_\lambda(x,y))/\epsilon} d\nu(y),$$

and

$$a' = \int h(y) e^{(\psi_2(y) - c_\lambda(x,y))/\epsilon} d\nu(y), \quad b' = \int e^{(\psi_2(y) - c_\lambda(x,y))/\epsilon} d\nu(y).$$

If

$$\|\psi_i\|_\infty \leq M,$$

then

$$b, b' \geq e^{-(M+2\lambda)/\epsilon} =: m_M > 0.$$

Using

$$\left| \frac{a}{b} - \frac{a'}{b'} \right| \leq \frac{|a - a'|}{b} + \frac{a'|b - b'|}{bb'},$$

we obtain

$$|D\mathcal{F}_\nu(\psi_1)x - D\mathcal{F}_\nu(\psi_2)x| \leq C\|e^{\psi_1/\epsilon} - e^{\psi_2/\epsilon}\|_\infty.$$

Since the exponential map is locally Lipschitz on bounded subsets of $L^\infty(\mathcal{X})$,

$$\|e^{\psi_1/\epsilon} - e^{\psi_2/\epsilon}\|_\infty \leq C\|\psi_1 - \psi_2\|_\infty.$$

Therefore

$$\|D\mathcal{F}_\nu(\psi_1) - D\mathcal{F}_\nu(\psi_2)\|_{\mathcal{L}(L^\infty)} \leq C\|\psi_1 - \psi_2\|_\infty.$$

Thus the derivative is continuous in the operator norm, and $\mathcal{F}_\nu$ is continuously Fréchet differentiable.

The same calculation applies to the second component. Therefore, combining the two components, the map

$$\mathcal{F} : L^\infty(\mathcal{X}) \times L^\infty(\mathcal{X}) \longrightarrow L^\infty(\mathcal{X}) \times L^\infty(\mathcal{X})$$

is continuously Fréchet differentiable.

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